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10,000-psi multilateral fracking system with large internal diameters for unconventional market

Abstract

Provided is a frac window system, a well system, and a wellbore stimulation method. The frac window system, in one aspect, includes an elongated tubular having a first end and a second end with an opening defined in a wall of the elongated tubular between the first end and the second end, the wall having an inner surface and an outer surface, wherein the opening in the wall is configured to align with a window of a wellbore casing. The frac window system, according to this aspect, may further include a polished bore receptacle coupled to the first end of the elongated tubular, the polished bore receptacle having an inside diameter (ID.sub.1) sufficient to engage with a high-pressure frac string.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5509483	12/1995	Graen	N/A	N/A
5992524	12/1998	Graham	N/A	N/A
6009949	12/1999	Gano	166/117.6	E21B 41/0042
6029747	12/1999	Morrell et al.	N/A	N/A
6263968	12/2000	Freeman	166/207	E21B 41/0042
6439312	12/2001	Hess et al.	N/A	N/A
6543541	12/2002	Buyers et al.	N/A	N/A
7213654	12/2006	Pluchek et al.	N/A	N/A
9574428	12/2016	Ramey, Jr. et al.	N/A	N/A
10435993	12/2018	Steele	N/A	E21B 41/0035
2012/0261130	12/2011	Linn et al.	N/A	N/A
2014/0144648	12/2013	McKay	N/A	N/A
2014/0151052	12/2013	Themig et al.	N/A	N/A
2018/0252053	12/2017	Rico	N/A	E21B 43/10
2018/0252075	12/2017	Dietz	N/A	N/A
2018/0283140	12/2017	Steele	N/A	N/A
2019/0010786	12/2018	Durst	N/A	E21B 41/0042
2022/0389792	12/2021	Steele	N/A	N/A
2022/0389795	12/2021	Steele	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2013043663	12/2012	WO	N/A
2016024088	12/2015	WO	N/A
2018222197	12/2017	WO	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims priority to U.S. Application Ser. No. 63/197,886, filed on Jun. 7, 2021, entitled “12,000-PSI MULTILATERAL FRACKING SYSTEM WITH LARGE INTERNAL DIAMETERS FOR UNCONVENTIONAL MARKET (AKA DEVICE, SYSTEM AND METHOD FOR FRACKING HIGH-PRESSURE WELLS WITHOUT A DRILLING RIG),” U.S. Application Ser. No. 63/197,945, filed on Jun. 7, 2021, entitled “MULTILATERAL WELL TOOLS THAT PASS THROUGH SMALLER RESTRICTIONS,” and U.S. Application Ser. No. 63/197,924, filed on Jun. 7, 2021, entitled “SPACER WINDOW SLEEVE,” all of which are commonly assigned with this application and incorporated herein by reference in their entirety.

BACKGROUND

- (1) In the production of hydrocarbons, it is common to drill one or more secondary wellbores from a first wellbore. Typically, the first and secondary wellbores, collectively referred to as a multilateral wellbore, will be drilled and/or cased using a drilling rig. Thereafter, once completed, the drilling rig will be removed, and the wellbores will produce hydrocarbons.
- (2) During any stage of the life of a wellbore, various treatment fluids may be used to stimulate the wellbore. As used herein, the term “treatment,” or “treating,” refers to any subterranean operation that uses a fluid in conjunction with a desired function and/or for a desired purpose. The term “treatment,” or “treating,” does not imply any particular action by the fluid or any particular component of the fluid.
- (3) One common stimulation operation that employs a treatment fluid is hydraulic fracturing. Hydraulic fracturing operations generally involve pumping a treatment fluid (e.g., a fracturing fluid) into a wellbore that penetrates a subterranean formation at a sufficient hydraulic pressure to create one or more cracks, or “fractures,” in the subterranean formation through which hydrocarbons will flow more freely. In some cases, hydraulic fracturing can be used to enhance one or more existing fractures. “Enhancing” one or more fractures in a subterranean formation, as that term is used herein, is defined to include the extension or enlargement of one or more natural or previously created fractures in the subterranean formation. “Enhancing” may also include positioning material (e.g., proppant) in the fractures to support (“prop”) them open after the hydraulic fracturing pressure has been decreased (or removed).
- (4) During the initial production life of a wellbore—often called the primary phase—primary production of hydrocarbons typically occurs either under natural pressure, or by means of pumps that are deployed within the wellbore. This may include wellbores that have undergone stimulation operations, such a hydraulic fracturing, during a completion process. Unconventional wells typically will not produce economical amounts of oil or gas unless they are stimulated via a hydraulic fracturing process to enhance and connect existing fractures. In order to reduce well costs, the hydraulic fracturing process is performed after the drilling rig has been removed from the well. Furthermore, wells may be hydraulically fractured without the aid of a workover rig if the equipment used to fracture a well is light enough to be transported in and out of the wellbore via a coiled tubing unit, wireline, electric line, or other device.
- (5) Over the life of a wellbore, the natural driving pressure may decrease to a point where the natural pressure is insufficient to drive the hydrocarbons to the surface given the natural permeability and fluid conductivity of the formation. At this point, the reservoir permeability and/or pressure must be enhanced by external means. In secondary recovery, treatment fluids are injected into the reservoir to supplement the natural permeability. Such treatment fluids may include water, natural gas, air, carbon dioxide or other gas and a proppant to hold the fractures open.
- (6) Likewise, in addition to enhancing the natural permeability of the reservoir, it is also common through tertiary recovery, to increase the mobility of the hydrocarbons themselves in order to enhance extraction, again through the use of treatment fluids. Such methods may include steam

injection, surfactant injection and carbon dioxide flooding.

(7) In both secondary and tertiary recovery, hydraulic fracturing may also be used to enhance production.

(8) Depending on the nature of the secondary or tertiary operation, it may be necessary to redeploy a rig, often referred to as a “workover rig,” to the wellbore to assist in these operations, which may require additional equipment be installed in a wellbore. For example, subjecting a producing wellbore to hydraulic fracturing pressures after it has been producing may damage certain casings, installations, or equipment already in a wellbore. Thus, it may be necessary to install additional equipment to protect the various equipment and tools already in the wellbore before proceeding with such operations. Such additional equipment is typically of sufficient size and weight that requires the use of a workover rig. As the number of secondary wellbores in a multilateral wellbore increases, the difficulty in protecting the various equipment in the first wellbore and the secondary wellbores becomes even more pronounced.

(9) It would be desirable to provide a system that avoids the need for drilling or workover rigs in treatment fluid operations in multilateral wellbores, particularly those subject to stimulation techniques such as hydraulic fracturing.

Description

BRIEF DESCRIPTION

(1) Reference is now made to the following descriptions taken in conjunction with the accompanying drawings, in which:

(2) FIGS. 1 and 2 illustrate a well system designed, manufactured and/or operated according to one or more embodiments of the disclosure;

(3) FIGS. 3 through 22 illustrate various different views of one embodiment of a well system, and use therefore, having large internal diameters according to one or more aspects of the present disclosure;

(4) FIGS. 23 through 39 illustrate various different views of one embodiment of a well system, and use therefore, employing high-expansion seals and/or high-expansion members according to one or more aspects of the present disclosure; and

(5) FIGS. 40A through 57 illustrate various different views of one embodiment of a well system, and use therefore, employing a spacer window sleeve according to one or more aspects of the present disclosure.

DETAILED DESCRIPTION

(6) The disclosure may repeat reference numerals and/or letters in the various examples or FIGs. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed. Further, spatially relative terms, such as beneath, below, lower, above, upper, uphole, downhole, upstream, downstream, and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated, the upward direction being toward the top of the corresponding FIG. and the downward direction being toward the bottom of the corresponding FIG., the uphole direction being toward the surface of the wellbore, the downhole direction being toward the toe of the wellbore. Unless otherwise stated, the spatially relative terms are intended to encompass different orientations of the apparatus in use or operation in addition to the orientation depicted in the FIGs. For example, if an apparatus in the FIGs. is turned over, elements described as being “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the exemplary term “below” can encompass both an orientation of above and below. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may

likewise be interpreted accordingly.

(7) Moreover, even though a FIG. may depict a horizontal wellbore or a vertical wellbore, unless indicated otherwise, it should be understood by those skilled in the art that the apparatus according to the present disclosure is equally well suited for use in wellbores having other orientations including vertical wellbores, deviated wellbores, multilateral wellbores, or the like. Likewise, unless otherwise noted, even though a FIG. may depict an offshore operation, it should be understood by those skilled in the art that the apparatus according to the present disclosure is equally well suited for use in onshore operations and vice-versa. Further, unless otherwise noted, even though a FIG. may depict a cased hole, it should be understood by those skilled in the art that the apparatus according to the present disclosure is equally well suited for use in open hole operations.

(8) As used herein, “first wellbore” shall mean a wellbore from which another wellbore extends (or is desired to be drilled, as the case may be). Likewise, a “second” or “secondary” wellbore shall mean a wellbore extending from another wellbore. The first wellbore may be a primary, main or parent wellbore, in which case, the secondary wellbore is a lateral or branch wellbore. In other instances, the first wellbore may be a lateral or branch wellbore, in which case the secondary wellbore is a “twig” or a “tertiary” wellbore.

(9) Generally, in one or more embodiments, a frac window system is provided in a multilateral wellbore with a secondary wellbore extending from a first wellbore. The frac window system includes a tubular having an opening therein that aligns with a secondary wellbore window formed in the casing string of the first wellbore. The frac window system may include annular seals along the outer surface of the tubular above and below the opening, and may further include an orientation device carried within the tubular.

(10) In one or more embodiments, an isolation sleeve (e.g., a main bore isolation sleeve) is positioned within the frac window system to seal the opening in the frac window system and the secondary wellbore window in the first wellbore casing to isolate the secondary wellbore from high pressure fluid directed farther down the first wellbore casing.

(11) In one or more alternative embodiments, a whipstock may seat on an orientation device so that a surface of the whipstock is aligned with the secondary wellbore window of the first wellbore casing string. In one or more embodiments, a straddle stimulation tool abuts the surface of the whipstock and extends through the frac window system opening from the first wellbore into the secondary wellbore, thereby providing the high pressure fluid to the secondary wellbore. In one or more embodiments, a plug may also be used to isolate the primary wellbore from the high pressure fluid directed to the secondary wellbore.

(12) Turning to FIGS. **1** and **2**, shown is an elevation view in partial cross-section of a frac window system **226** deployed in a well system **10** (land based in FIG. **1** and offshore in FIG. **2**) utilized to produce hydrocarbons from wellbore **12** extending through various earth strata in a petroleum formation **14** located below the earth's surface **16**. Wellbore **12** may be formed of a single first wellbore and may include one or more second or secondary wellbores **12a**, **12b** . . . **12n**, extending into the formation **14**, and disposed in any orientation and spacing, such as the horizontal secondary wellbores **12a**, **12b** illustrated.

(13) Well system **10** includes a drilling rig or derrick **20**. Drilling rig **20** may include a hoisting apparatus **22**, a travel block **24**, and a swivel **26** for raising and lowering a conveyance such as tubing string **30**. Other types of conveyances may include tubulars such as casing, drill pipe, coiled tubing, production tubing, and other types of pipe or tubing strings. Still other types of conveyances may include wirelines, slicklines, and the like. In FIG. **1**, tubing string **30** is a substantially tubular, axially extending work string formed of a plurality of drill pipe joints coupled together end-to-end, while in FIG. **2**, tubing string **30** is completion tubing supporting a completion assembly as described below. Drilling rig **12** may include a kelly **32**, a rotary table **34**, and other equipment associated with rotation and/or translation of tubing string **30** within a wellbore **12**. For

some applications, drilling rig **20** may also include a top drive unit **36**.

(14) Drilling rig **20** may be located proximate to a wellhead **40** as shown in FIG. **1**, or spaced apart from wellhead **40**, such as in the case of an offshore arrangement as shown in FIG. **2**. One or more pressure control devices **42**, such as blowout preventers (BOPs) and other equipment associated with drilling or producing a wellbore may also be provided at wellhead **40** or elsewhere in the well system **10**.

(15) For offshore operations, as shown in FIG. **2**, whether drilling or production, drilling rig **20** may be mounted on an oil or gas platform, such as the offshore platform **44** as illustrated, or on semi-submersibles, drill ships, and the like (not shown). Well system **10** of FIG. **2** is illustrated as being a marine-based production system. Likewise, well system **10** of FIG. **1** is illustrated as being a land-based production system. In any event, for marine-based systems, one or more subsea conduits or risers **46** extend from deck **50** of platform **44** to a subsea wellhead **40**. Tubing string **30** extends down from drilling rig **20**, through riser **46** and BOP **42** into wellbore **12**.

(16) A fluid source **52**, such as a storage tank or vessel, may supply a working or service fluid **54** pumped to the upper end of tubing string **30** and flow through tubing string **30**. Fluid source **52** may supply any fluid utilized in wellbore operations, including without limitation, drilling fluid, cementitious slurry, acidizing fluid, liquid water, steam, hydraulic fracturing fluid or some other type of fluid.

(17) Wellbore **12** may include subsurface equipment **56** disposed therein, such as, for example, the completion equipment illustrated in FIG. **1** or **2**. In other embodiments, the subsurface equipment **56** may include a drill bit and bottom hole assembly (BHA), a work string with tools carried on the work string, a completion string and completion equipment or some other type of wellbore tool or equipment.

(18) Well system **10** may generally be characterized as having a pipe system **58**. For purposes of this disclosure, pipe system **58** may include casing, risers, tubing, drill strings, completion or production strings, subs, heads or any other pipes, tubes or equipment that attaches to the foregoing, such as tubing string **30** and riser **46**, as well as the wellbore and laterals in which the pipes, casing and strings may be deployed. In this regard, pipe system **58** may include one or more casing strings **60** that may be cemented in wellbore **12**, such as the surface, intermediate and production casing strings **60** shown in FIG. **1**. An annulus **62** is formed between the walls of sets of adjacent tubular components, such as concentric casing strings **60** or the exterior of tubing string **30** and the inside wall of wellbore **12** or casing string **60**, as the case may be.

(19) As shown in FIGS. **1** and **2**, where subsurface equipment **56** is illustrated as completion equipment, disposed in secondary wellbore **12a** is a lower completion assembly **82** that includes various tools such as an orientation and alignment subassembly **84**, a packer **86**, a sand control screen assembly **88**, a packer **90**, a sand control screen assembly **92**, a packer **94**, a sand control screen assembly **96** and a packer **98**.

(20) Extending uphole and downhole from lower completion assembly **82** is one or more communication cables **100**, such as a sensor or electric cable, that passes through packers **86**, **90** and **94** and is operably associated with one or more electrical devices **102** associated with lower completion assembly **82**, such as sensors positioned adjacent sand control screen assemblies **88**, **92**, **96** or at the sand face of formation **14**, or downhole controllers or actuators used to operate downhole tools or fluid flow control devices. Cable **100** may operate as communication media, to transmit power, or data and the like between lower completion assembly **82** and an upper completion assembly **104**.

(21) In this regard, disposed in wellbore **12**, the upper completion assembly **104** is coupled at the lower end of tubing string **30**. The upper completion assembly **104** includes various tools such as a packer **106**, an expansion joint **108**, a packer **110**, a fluid flow control module **112** and an anchor assembly **114**.

(22) Extending uphole from upper completion assembly **104** are one or more communication

cables **116**, such as a sensor cable or an electric cable, which passes through packers **106**, **110** and extends to the surface **16**. Cable(s) **116** may operate as communication media, to transmit power, or data and the like between a surface controller (not pictured) and the upper and lower completion assemblies **104**, **82**.

(23) Fluids, cuttings, and other debris returning to surface **16** from wellbore **12** may be directed by a flow line **118** back to storage tanks, fluid source **52** and/or processing systems **120**, such as shakers, centrifuges, and the like.

(24) In each of FIGS. **1** and **2**, a frac window system **226** is generally illustrated. Frac window system **226** is positioned adjacent secondary wellbore **12b** so that an opening **132** in the frac window system **226** is aligned with the casing window **134** of casing string **60** adjacent secondary wellbore **12b**.

(25) FIG. **3** is an elevation view in cross-section of the first wellbore **12** and the upper and lower secondary wellbores, **12b** and **12a**, respectively, illustrated as extending from first wellbore **12** in more detail. Specifically, the first wellbore **12** is illustrated as being at least partially cased with a first wellbore casing **200** cemented therein. While generally illustrated as vertical, first wellbore **12**, as well as any of the wellbores described, may have any orientation. In any event, at the distal end **202** of first wellbore **12**, a casing hanger **204** may be deployed from which a secondary wellbore casing **206** hangs. Secondary wellbore casing **206** has a proximal end **206a** and a distal end **206b**. The proximal end **206a** may include a shoulder **208** for supporting secondary wellbore casing **206** on hanger **204**. The distal end **206b** may include perforations **207** or sliding sleeves. Secondary wellbore casing **206** is illustrated as cemented in place within wellbore **12a**. Proximal end **206a** may also include a polished bore receptacle (PBR) **215**, which may be positioned above liner hanger **204**. PBR **215** may have a larger inner diameter than the secondary wellbore casing **206**. This prevents a seal **242** (see FIG. **4A**) from creating a restriction smaller than the casing **206** inner diameter.

(26) Likewise, with regard to secondary wellbore **12b**, which is formed at a junction **209** with first wellbore **12**, a transition joint **210** extends from a casing window **212** formed along the inner annulus of casing **200**. Transition joint **210** may be made of steel, fiberglass, or any material capable of supporting itself under the pressure of fluids, cement, or solid objects such as rock in a downhole environment. A casing hanger **214** may be deployed from which a secondary wellbore casing **216** hangs. Secondary wellbore casing **216** has a proximal end **216a** and a distal end **216b** and an interior surface **216i**. The distal end **216b** may include perforations **217** or sliding sleeves. The proximal end **216a** may include a shoulder **218** for supporting casing **216** on hanger **214**. Secondary wellbore casing **216** is illustrated as cemented in place within wellbore **12b**. In other embodiments (not shown) the transition joint **210** may be threaded directly to a PBR, which in turn is threaded to the secondary wellbore casing **216**, and no casing hanger **214** is necessary.

(27) Persons of ordinary skill in the art will appreciate that the illustrated first wellbore **12** and secondary wellbores **12a**, **12b**, and the equipment illustrated therein, are for illustrative purposes only, and are not intended to be limiting. For example, secondary wellbore casing strings **206**, **216** are not limited to a particular size or manner of support, and other systems for supporting secondary wellbore casing may be utilized.

(28) Any one or more of the casing strings or tubulars described herein may include an engagement mechanism **220** deployed along an inner surface and disposed to engage a cooperating engagement mechanism, such as engagement mechanism **246** (FIG. **4A**) described below, to secure or otherwise anchor adjacent tubulars relative to one another at a desired depth and/or orientation. In one or more embodiments, engagement mechanism **220** may be latch couplings as are shown deployed along first wellbore casing **200**. In one or more embodiments, an engagement mechanism **220** is positioned adjacent to window **212** at a known distance. In one or more embodiments, an engagement mechanism **220** is positioned adjacent window **212** upstream or above junction **209**, while in other embodiments, the engagement mechanism is positioned adjacent window **212**

downstream or below junction **209**. The disclosure is not limited to a particular type of engagement mechanism **220**.

(29) Similar to engagement mechanism **220**, an engagement mechanism **222** is illustrated along the interior surface **216i** of casing **216**.

(30) Turning to FIG. **4A**, an elevation view in cross section illustrates the frac window system **226** deployed adjacent junction **209** within first wellbore casing **200**. Frac window system **226** is formed of an elongated tubular **228** having a first end **228a** and a second end **228b** with an opening **230** defined in a wall **232** of the tubular between ends **228a**, **228b**. The elongated tubular **228** may extend a significant distance, and may be constructed of multiple casing, tubing, or other pipe without departing from the scope and spirit of the disclosure. Elongated tubular **228** includes an inner surface **234** and an outer surface **236**.

(31) An orientation device **238** is disposed or otherwise formed along the inner surface **234** of elongated tubular **228**. In one or more embodiments, orientation device **238** is located below the opening **230**, between opening the **230** and the second end **228b** of elongated tubular **228**. Although orientation device **238** may be any mechanism or device that permits rotational orientation of a tool or equipment within elongated tubular **228**, in one or more embodiments, orientation device **238** may be a scoop head, a muleshoe or a ramped or angled surface. In yet another embodiment, the orientation device **238** is located above the opening **230**.

(32) Frac window system **226** further includes a first seal **240** disposed along the outer surface **236** of the elongated tubular **228**. In one or more embodiments, first seal **240** is disposed along the outer surface **236** between the opening **230** and the first end **228a** of the elongated tubular **228**. Likewise, a second seal **242** is disposed along the outer surface **236** below opening **230** between opening **230** and the second end **228b** of elongated tubular **228**. First seal **240** extends between frac window **226** and casing **200** to seal the annular space **244** therebetween. Likewise, second seal **242** extends between the outer surface **236** of the elongated tubular **228** and an inner surface of the adjacent tubular, e.g., first wellbore casing **200**, to seal the annular space about the second end **228b** of elongated tubular **228**. In the illustrated embodiment, second end **228b** extends into proximal end **206a** of secondary wellbore casing **206**, and in such case, second seal **242** seals the annular space therebetween. In other embodiments, second seal **242** may be disposed along the end of **228b** of elongated tubular **228** to seal between frac window system **226** and the first wellbore casing **200**, and in particular, in some embodiments, PBR **215**. In other embodiments, second seal **242** may be disposed along the inner surface **234** of the elongated tubular **228** at the second end of **228b** to seal between frac window system **226** and a tubular (not shown) extending therein.

(33) Seals **240**, **242** as described may be any mechanism that can seal an annular space between tubulars, such as for example an expandable liner hanger system, swellable elastomer or otherwise, any type of, or combination of, elastomeric element(s) or composite elements made of man-made and/or natural materials that may be deployed to effectuate a sealing contact with both tubulars as described. A seal may include a shoulder, such as shoulder **252** formed along the outer surface **236** of elongated tubular **228**. The elongated tubular **228** may include a plurality of joints of pipe spanning the distance between the shoulder **252** and smooth sealing surfaces **254** may also be provided along the inner surface **234** of the elongated tubular **228**. The shoulder **252** may engage a similarly formed shoulder, such as the end of secondary wellbore casing **206**, against which shoulder **252** may seat, forming a metal-to-metal seal. Although not limited to a particular configuration, the most common place shoulder **252** would engage is in the PBR **215** attached to hanger **204**. This would typically be an “anchor” type of mechanism wherein shoulder **252** would have a releasable anchoring device such as a latch, a lug, a snap or similar mechanism, to attach itself to the top of the PBR **215** or to the top of hanger **204**. The top of PBR **215** or the top of hanger **204** may include a receiving head, a lug-receiver, a snap locator, or other device to receive, releasably secure, and/or provide a sealing surface for shoulder **252**, and/or seal **242** and/or end **228b** of elongated tubular **228**. The disclosure is not limited to a particular type of mechanism that

can seal an annular space between tubulars.

(34) In other embodiments, shoulder **252** may be disposed along the inner surface **234** of end of **228b** of elongated tubular **228** to engage a similarly formed shoulder, such as the end of secondary wellbore casing **206**.

(35) Frac window system **226** may further include an engagement mechanism **246** along outer surface **236** and disposed for engagement with an engagement mechanism **220**. In one or more embodiments, engagement mechanism **246** is a latch and engagement mechanism **220** is a latch coupling.

(36) In one or more embodiments, engagement mechanism **246** may be an Engagement.

Orientation, and Depth (EOD) device that provides depth, orientation, and an engagement into an accepting device. The engagement device of the EOD may be one that is releasable. The EOD may provide depth, orientation, and releasable engagement in concert with a device such as engagement mechanism **220** or engagement mechanism **222** or against a surface of a pipe or other device having a generally circular form and an inner and outer surface. In further embodiments, engagement mechanism **246** may be a collet. In other embodiments, engagement mechanism **246** may be a multiplicity of collets, keys, slips, latches, etc. Engagement mechanism **246** may also consist of multiple devices to provide depth, orientation and/or engagement such as collets, keys, slips, and/or latches, etc. Thus, for example, the engagement mechanism **246** in the form of an EOD may be mounted on the outer surface **236** of the elongated tubular **228** for engagement with an engagement mechanism **220**, such as a latch coupling, disposed along the interior annulus of the first wellbore casing **200**. In one or more embodiments, the engagement mechanism **220** of the casing **200** is above window **212**, and the EOD **246** of frac window system **226** is between the opening **230** and first end **228a** of the tubular. In one or more embodiments, the EOD **246** is between the first seal **240** and the first end **228a** of the tubular. It will be appreciated that in one or more embodiments, engagement mechanism **246** may function to releasably engage another engagement mechanism, such as engagement mechanism **220** or **222**; function as a no-go shoulder (depth lock or stop) at a desired depth; and provide an orientation lock at a desired orientation.

(37) In any event, regardless of the particular type, in one or more embodiments, although engagement mechanism **246** may be disposed anywhere along the outer surface **236** so long as the axial position between frac window system **226** and window **212** is established, engagement mechanism **246** is disposed between the opening **230** and the first end **228a** to engage an engagement mechanism **220** upstream of window **212**, as illustrated. In one or more embodiments, the engagement mechanism **246** is between the first seal **240** and the first end **228a** so that the engagement mechanism **246** may be isolated from pressurized fluid that may be introduced into one of the secondary wellbores **12a**, **12b**. In another embodiment, the latch **246** is placed below the window opening **230** (e.g., a downhole end of the window opening) and the distal end **228b**.

(38) As will be appreciated, when engagement mechanism **246** is a latch and engagement mechanism **220** is a latch coupling, cooperation between the two mechanism **220**, **246** can be utilized to both axially and radially position frac window system **226**. However, in one or more embodiments, engagement mechanism **220** need not be present. Rather, engagement mechanism **246** may be another type of device or mechanism to secure and/or position frac window system **226** in wellbore **12**. In one or more embodiments, engagement mechanism **246** may be an expandable liner hanger carried on the outer surface **236** of elongated tubular **228**. Alternatively, or in addition, engagement mechanism **246** may be one or more slips that can be actuated to anchor against the first wellbore casing (or the wall of first wellbore **12** in the instance of an uncased wellbore). In one or more embodiments, engagement mechanism **246** may be one or more collets. In other embodiments, **246** may be a multiplicity of collets, keys, slips, latches, pockets, grooves, recesses, indentations, slots, splines, etc. Also, mechanism **220** may consist of multiple devices to provide depth, orientation and/or engagement such as collets, keys, slips, and/or latches, etc. The disclosure is not limited to a particular type of engagement mechanism. Alternatively, or in addition, in one or

more embodiments, engagement mechanism **246** may be, or work in concert with, a mechanically, hydraulically, and/or electrically activated window finder deployed within elongated tubular **228** that will actuate and extend at least partially through opening **230** and window **212** when the opening **230** and casing window **212** are aligned. In such case, it will be appreciated, with the relative alignment achieved, another engagement mechanism, such as an expandable liner hanger or slips, may be actuated to anchor elongated tubular **228** in position.

(39) It will be appreciated that latch **246** and latch coupling **220** permit frac window system **226** to be axially and radially oriented so that frac window system **226** is adjacent junction **209**, and thus window **212**, and that opening **230** is aligned with window **212** of casing **200**.

(40) Frac window system **226** may further include a first depth mechanism **248** disposed along the inner surface **234**. In one or more embodiments, the first depth mechanism **248** is between the opening **230** and the first end **228a** of elongated tubular **228**. Similarly, a depth mechanism **250** may be disposed along the inner surface **234** adjacent the orientation device **238**.

(41) When deployed as described above, opening **230** of frac window system **226** is aligned with window **212** of casing **200** and the annulus about elongated tubular **228** is sealed above and below window **212**. In one or more embodiments, opening **230** of frac window system **226** has a dimension $L_{sub.1}$ that is smaller than the dimension L_2 of window **212**.

(42) One or more of the inner or outer surfaces of elongated tubular **228** adjacent the ends **228a**, **228b** may be threaded to assist in deployment of elongated tubular **228**. For example, the inner surface **234** of elongated tubular **228** adjacent first end **228a** may be threaded while the inner surface **234** adjacent second end **228b**, as well as the outer surface **236** adjacent the two ends **228a**, **228b** may be smooth, the threads disposed to permit attachment of a running tool (not shown). However, in one or more embodiments, the inner and outer surfaces **234**, **236** adjacent the ends **228a**, **228b** are all sufficiently smooth to permit an elastomeric element to seal against the surface. Thus, as used herein, “smooth” is used to refer to a surface that is not threaded. The smooth surface may have other shapes, features or contours, but is not otherwise disposed to engage the threads of another mechanism in order to join the mechanism to the surface. Other smooth sealing surfaces **254** may also be provided along the inner surface **234** of the elongated tubular **228** to ensure a desired level of sealing during operations employing frac window system **226**.

(43) The present disclosure has recognized that one of the roadblocks from fracking multilateral wells is the necessity of utilizing a drilling rig during the fracking operations. A frac window system designed, manufactured and operated according to the novel aspects of this disclosure, such as the frac window system **226**, allows the drilling rig to be moved off of the well and coiled tubing to be utilized for substantially all (or all) frac operations. For example, a frac window system designed, manufactured and/or operated according to the novel aspects of this disclosure allows high rate, high-pressure through workstring/production tubing. For example, a frac window system designed, manufactured and/or operated according to the novel aspects of this disclosure may be capable of withstanding the high-pressures and stresses of stimulating at pressures of at least 5,000-psi, at least 10,000-psi, at least 12,500-psi and/or at least 15,000-psi. In some highly specialized situations, it may be desirable to be able to withstand pressures of 30,000-psi or more. For example, fracking of >80 BPM at 12,500-psi is achievable.

(44) In some embodiments, it may be desirable to use wireline tools to perforate a portion of a wellbore (e.g., **207** in liner **206** in wellbore **12a** and/or **217** in **216** in wellbore **12b**). As a rule of thumb, wireline tools (electric line, slick line, braided line, cable line, sand line, etc.) cannot move downhole solely due to gravity in wellbores with an inclination of greater than 65-degrees. For that reason, devices may be attached to the wireline so the wireline can be pumped downhole—fluid pressure is applied at the surface to pump the device and wireline downhole.

(45) Accordingly, a frac window system designed, manufactured and/or operated according to the novel aspects of this disclosure may be manufactured of certain materials, and may have certain sidewall thicknesses and inside diameters, which would allow it to withstand the foregoing high-

pressures. For example, wherein traditional frac window systems might comprise lower cost low alloy steels (e.g., L-80 material, K-55 material, etc.) that are only rated up to 5,000-psi, a frac window system according to the present disclosure would comprise high-strength materials that are greater than 5,000-psi rated, if not greater than 10,000-psi rated, if not greater than 12,500-psi rated, if not greater than 15,000-psi rated, or even up to 30,000-psi rated. In at least one embodiment, a frac window system according to the disclosure includes materials having a minimum yield strength of at least 110-ksi, if not at least 125-ksi, if not at least 140-ksi, among others. For example, a Q125 steel could be used for at least a portion of the frac window system and remain within the scope of the disclosure.

(46) Additionally, a frac window system designed, manufactured and/or operated according to the novel aspects of the disclosure could include an enlarged upper polished bore receptacle (PBR). The enlarged PBR, in at least one embodiment, would have an inside diameter (ID.sub.1) sufficient to engage with a high-pressure frac string. The term “high-pressure frac string”, as used herein, is defined as a frac string capable of providing frac pressures of at least 5,000-psi (e.g., 340 atm). In at least one other embodiment, the enlarged PBR would have an inside diameter (ID.sub.1) sufficient to engage with a high-pressure frac string. The term “extremely high-pressure frac string”, as used herein, is defined as a frac string capable of providing frac pressures of at least 10,000-psi (e.g., 680 atm). In yet at least one other embodiment, the enlarged PBR would have an inside diameter (ID.sub.1) sufficient to engage with an extremely high-pressure frac string. The term “super high-pressure frac string”, as used herein, is defined as a frac string capable of providing frac pressures of at least 12,500-psi (e.g., 851 atm). In yet at least one other embodiment, the enlarged PBR would have an inside diameter (ID.sub.1) sufficient to engage with a super high-pressure frac string. Nevertheless, in even yet other embodiments, the enlarged PBR, in at least one embodiment, would have an inside diameter (ID.sub.1) sufficient to engage with a frac string capable of providing frac pressures of at least 15,000-psi (e.g., 1021 atm), if not at least 30,000-psi (e.g., 2041 atm). While the present disclosure is discussed mainly with regard to a high-pressure frac string, other embodiments wherein extremely high-pressure and super high-pressure frac strings are used are within the scope of the disclosure. In at least one embodiment, the enlarged PBR has an inside diameter (ID.sub.1) of at least 5" (e.g., 12.7 cm), if not at least 5.5" (e.g., 13.97 cm), if not at least 6" (e.g., 15.24 cm), if not at least 6.5" (e.g., 16.51 cm), if not at least 7" (e.g., 17.78 cm), or more. In at least one embodiment, the enlarged PBR includes an outside diameter (OD.sub.1) of at least 8.2" (e.g., 20.83 cm) and an inside diameter (ID.sub.1) of at least 7.1" (e.g., 18.03 cm). Furthermore, such an enlarged PBR could include at least 110-ksi grade material, if not at least 125-ksi grade material capable of handling an internal yield pressure of at least 14,850-psi (e.g., 1010 atm) (1.25 S.F. at 200 deg. F.).

(47) The enlarged PBR, accordingly, would allow a larger high-pressure frac string to engage therewith and deploy larger wellbore features within an existing multilateral wellbore. For example, the enlarged PBR has the ability to pass large frac plugs through the system during frac operations (e.g., at least 5.5" (e.g., 12.7 cm) frac plugs in 9⁵/₈" (e.g., 25.45 cm) well), and has the ability to frac more than one lateral (wellbore) sequentially with minimal trips (and no drilling rig). The enlarged PBR also has the ability to pass plugs therethrough (e.g., plug **274** of FIG. **6A**), whipstocks therethrough (e.g., whipstock **276** of FIG. **7**), straddle stimulation tools therethrough (e.g., straddle stimulation tool **285** of FIG. **8**), lateral bore frac plugs therethrough (e.g., frac plugs **910** of FIG. **9A**), main bore isolation sleeve therethrough (e.g., main bore isolation sleeve **1560** of FIG. **15A**), main bore frac plugs therethrough (e.g., frac plugs **1610** of FIG. **16**), etc.

(48) A frac window system designed, manufactured and/or operated according to the novel aspects of the disclosure could be left in the well as completion equipment or the equipment may be utilized as a service tool. In those situations wherein the novel frac window system is employed as a service tool, the novel frac window system could be re-dressed (e.g., replace seals) and used multiple times. If the novel frac window system is made of high-strength, e.g., CRA material, it

will be more erosion-resistant and capable of being used on perhaps as many as 50 jobs with the only cost being to replace seals and the most highly erodible pieces (e.g., seal nipples, the tubing used in the Lateral Isolation Sleeve, etc.).

(49) Given the foregoing, the frac window system **226** of FIG. **4A** may include, in at least one embodiment, an enlarged PBR **255**. Again, the enlarged PBR may have a sufficient inside diameter (ID.sub.1) to engage with a high-pressure frac string. Furthermore, the enlarged PBR **255** may have a sufficient inside diameter (ID.sub.1) to pass the larger wellbore features discussed in the paragraphs above. Thus, in at least one embodiment, the enlarged PBR has an inside diameter (ID.sub.1) of at least 5" (e.g., 12.7 cm), if not at least 5.5" (e.g., 13.97 cm), if not at least 6" (e.g., 15.24 cm), if not at least 6.5" (e.g., 16.51 cm), if not at least 7" (e.g., 17.78 cm), or more. In at least one embodiment, the enlarged PBR includes an outside diameter (OD.sub.1) of at least 8.2" (e.g., 20.83 cm) and an inside diameter (ID.sub.1) of at least 7.1" (e.g., 18.03 cm).

(50) Turning to FIGS. **4B** and **4C**, illustrated is an isometric view and a cross-sectional view, respectively, of an alternative embodiment of a frac window system **400**, as might be used within a well system, such as the well system of FIGS. **1** and **2**. The frac window system **400**, in at least one embodiment, is similar to the frac window system **226** illustrated in FIG. **4A**. The frac window system **400**, in the illustrated embodiment, includes an enlarged PBR **410** (e.g., as might be used to seal/engage a larger diameter of a high-pressure frac string), a window exit **420** (e.g., as might be used to access a lateral wellbore), an inner orientation device **440** (e.g., muleshoe), an axial orientation device **460**, and a rotational orientation device **480**.

(51) Turning to FIG. **4D**, illustrated is an enlarged cross-sectional view of the PBR **410** of FIGS. **4B** and **4C**. In the illustrated embodiment of FIG. **4D**, the PBR **410** includes an inside diameter (ID.sub.1) and an outside diameter (OD.sub.1). The inside diameter (ID.sub.1), in at least one embodiment, is at least 5" (e.g., 12.7 cm), if not at least 5.5" (e.g., 13.97 cm), if not at least 6" (e.g., 15.24 cm), if not at least 6.5" (e.g., 16.51 cm), if not at least 7" (e.g., 17.78 cm), or more. Again, the inside diameter (ID.sub.1) may be based upon the outside diameter (OD.sub.5) of a high-pressure frac string that the PBR **410** is configured to engage with. In at least one embodiment, the PBR **410** includes an outside diameter (OD.sub.1) of at least 8.2" (e.g., 20.83 cm) and an inside diameter (ID.sub.1) of at least 7.1" (e.g., 18.03 cm), and furthermore comprises at least 125-ksi grade material that can accommodate an internal yield pressure of at least 14,850-psi (e.g., 1010 atm).

(52) In yet another embodiment, the PBR **410** has a wall thickness (t.sub.1) and a length (L.sub.1). In at least one embodiment, the wall thickness (t.sub.1) is at least 0.2" (e.g., 0.51 cm), if not at least 0.5" (e.g., 1.27 cm), if not at least 3" (e.g., 7.62 cm). In at least one embodiment, the wall thickness (t.sub.1) ranges from 0.3" (e.g., 0.76 cm) to 0.7" (e.g., 1.78 cm). In at least one embodiment, the length (L.sub.1) is at least 6" (e.g., 15.24 cm), if not at least 48" (e.g., 122 cm), if not at least 360" (e.g., 914 cm). In at least one embodiment, the length (L.sub.1) ranges from 36" (e.g., 91.4 cm) to 120" (e.g., 305 cm).

(53) Further illustrated in FIG. **4D** is an upper window nipple **412**, which would attach to the PBR **410**. The upper window nipple **412**, in at least one embodiment, might comprise at least 110-ksi CRA grade material that can accommodate an internal yield pressure of at least 12,500-psi (e.g., 851 atm). Furthermore, the upper window nipple **412** might have an inside diameter (ID.sub.2) of at least 5" (e.g., 12.7 cm), if not at least 5.7" (e.g., 14.48 cm). The upper window nipple **412**, in the illustrated embodiment, includes one or more profiles **414** for engaging with an isolation sleeve (not shown).

(54) Turning to FIG. **4E**, illustrated is an enlarged cross-sectional view of the window exit **420**. The window exit **420**, in at least one embodiment, might comprise at least 110-ksi CRA grade material that can accommodate an internal yield pressure of at least 12,500-psi (e.g., 851 atm). Furthermore, the window exit might have an inside diameter (ID.sub.3) of at least 5" (e.g., 12.7 cm), if not at least 5.7" (e.g., 14.48 cm).

(55) Turning to FIG. **4F**, illustrated is an enlarged cross-sectional view of a lower portion of the

window exit **420**, including a lower window nipple **422**. The lower window nipple **422**, in at least one embodiment, might comprise at least 110-ksi CRA grade material that can accommodate an internal yield pressure of at least 12,500-psi (e.g., 851 atm). Furthermore, the lower window nipple **422** might have an inside diameter (ID.sub.4) of at least 5" (e.g., 12.7 cm), if not at least 5.7" (e.g., 14.48 cm). The lower window nipple **422**, in the illustrated embodiment, may additionally include one or more profiles **424** for engaging with an isolation sleeve (not shown). FIG. **4F** additionally illustrates an inner orientation device **440** (e.g., muleshoe).

(56) Turning to FIG. **4G**, illustrated are the axial orientation device **460** and the rotational orientation device **480**. As those skilled in the art appreciate, the axial orientation device **460** and the rotational orientation device **480** may engage with wellbore casing to position the frac window system **400** at the appropriate location within the wellbore.

(57) Turning to FIG. **5**, the frac window system **226** is illustrated with a main bore isolation sleeve **260** deployed therein. For example, in certain embodiments the frac window system **226** is run in hole with the main bore isolation sleeve **260** disposed therein. Main bore isolation sleeve **260** is formed of a tubular sleeve **262** having a first end **262a** and a second end **262b**. Tubular sleeve **262** has an inner surface **264** and an outer surface **266**.

(58) Disposed along the outer surface **266** of tubular sleeve **262** are a first sleeve seal **268** and a second sleeve seal **270**. First and second sleeve seals **268**, **270** are spaced apart, as described below, to seal above and below opening **230** when main bore isolation sleeve **260** is deployed within frac window system **226**.

(59) Also disposed along the outer surface **266** of tubular sleeve **262** is a depth mechanism **272**. In one or more embodiments, depth mechanism **272** is positioned between the first sleeve seal **268** and the first end **262a**. Depth mechanism **272** is disposed to engage a depth mechanism disposed along the inner surface **234** of elongated tubular **228** of frac window system **226**. In the illustrated embodiment, sleeve depth mechanism **272** engages first depth mechanism **248** of frac window system **226**. When depth mechanism **272** is so engaged, the first end **262a** of tubular sleeve **262** is above the opening **230** in the elongated tubular **228** and the second end **262b** of tubular sleeve **262** is below the opening **230** in the elongated tubular **228** of frac window system **226**. Moreover, when depth mechanism **272** is so engaged, the first sleeve seal **268** of tubular sleeve **262** is above the opening **230** in the elongated tubular **228** and the second sleeve seal **270** of tubular sleeve **262** is below the opening **230** in the elongated tubular **228** of frac window system **226**, such that secondary wellbore **12b** is isolated from first wellbore **12**. In other words, fluid communication between secondary wellbore **12b** and first wellbore **12** is blocked by main bore isolation sleeve **260**, allowing various operations, such as high-pressure pumping, in the first wellbore **12** or secondary wellbore **12a** to occur without impacting secondary wellbore **12b**.

(60) In FIG. **6A**, a high-pressure frac string **251** is shown as disposed within the wellbore **12**. While a high-pressure frac string is illustrated, in other embodiments an extremely high-pressure frac string or super high-pressure frac string may be used. In the illustrated embodiment, the high-pressure frac string **251** is positioned within the PBR **255**. In at least one embodiment, the high-pressure frac string **251** includes an outside diameter (OD.sub.5) and an inside diameter (ID.sub.5). (See, FIG. **6B**) The outside diameter (OW, in at least one embodiment, is at least 5" (e.g., 12.7 cm), if not at least 5.5" (e.g., 13.97 cm), if not at least 6" (e.g., 15.24 cm), if not at least 6.5" (e.g., 16.51 cm), if not at least 7" (e.g., 17.78 cm), or more. Again, the outside diameter (OD.sub.5) may be based upon the inside diameter (ID.sub.1) of the PRB **255** it is configured to engage with. In at least one embodiment, the high-pressure frac string **251** includes an inside diameter (ID.sub.5) of at least 4.5" (e.g., 11.43 cm), if not at least 5" (e.g., 12.7 cm), if not at least 5.5" (e.g., 13.97 cm), if not at least 6" (e.g., 15.24 cm), if not at least 6.5" (e.g., 16.51 cm), or more. In yet another embodiment, the high-pressure frac string **251** has a wall thickness (t.sub.5). In at least one embodiment, the wall thickness (t.sub.5) of the high-pressure frac string **251** is at least 0.2" (e.g., 0.51 cm), if not at least 0.54" (e.g., 1.37 cm), if not at least 2.1" (e.g., 5.33 cm). In at least one

embodiment, the wall thickness (t.sub.5) of the high-pressure frac string **251** ranges from 0.5" (e.g., 1.27 cm) to 0.75" (e.g., 1.91 cm).

(61) The frac window system **226** is illustrated with a plug **274** deployed in the lower secondary wellbore **12a**. Much in the same way that main bore isolation sleeve **260** is utilized to isolate secondary wellbore **12b**, the plug **274** may be deployed to isolate secondary wellbore **12a** from pumping operations relating to secondary wellbore **12b**. Plug **274** may be set at any time. In some embodiments, plug **274** is set before running in frac window system **226**, while in other embodiments, plug **274** may be set on the same run-in trip as frac window system **226**, while in other embodiments, plug **274** may be run in and set after frac window system **226** is in place, for example through the high-pressure frac string **251**. Nevertheless, in at least one embodiment the plug **274** may be positioned within frac window system **226**, preferably at a location adjacent end **228b** or may be positioned in casing **206** of secondary wellbore **12a** or within PBR **215** (FIG. 5) if present.

(62) FIG. 6B illustrates a zoomed in view of the PBR **410** of FIGS. 4A through 4D, with a high-pressure frac string **651** disposed therein. In at least one embodiment, the high-pressure frac string **651** extends within the PBR **410** by a distance (d). The distance (d) may vary greatly and remain within the scope of the disclosure, but in at least one embodiment the distance (d) is at least 3" (e.g., 7.62 cm), if not at least 120" (e.g., 305 cm), if not at least 360" (e.g., 914 cm).

(63) In FIG. 7, a whipstock **276** is illustrated as deployed in frac window system **226**. Whipstock **276** may be of any shape or configuration, but generally has first end **278** and a second end **280** with a contoured surface **282** at first end **278**. Whipstock **276** may include a follower **281**, such as a lug or similar device. Follower **281** is preferably positioned along the outer surface **283** of whipstock **276** and may protrude from the surface **283** to engage orientation device **238** of frac window system **226** in order to rotate whipstock **276** to the desired angular position within first wellbore **12**. Likewise, whipstock **276** may include a depth mechanism **284** disposed to engage the mechanism **250** to secure the oriented whipstock **276** to elongated tubular **228** of frac window system **226**. More specifically, when whipstock **276** is deployed within frac window system **226**, whipstock **276** is axially positioned so that the first end **278** of whipstock **276** is adjacent opening **230** and radially positioned so that the contoured surface **282** will direct, deflect, or otherwise guide tools and other devices passing down through first wellbore **12** through opening **230** and into secondary wellbore **12b**.

(64) It should be appreciated that as described herein, whipstock **276** is not limited to any particular type of whipstock, but may be any device which will deflect, direct or otherwise guide a tool or device through opening **230**. In some embodiments, whipstock **276** may be a solid body, while in other embodiments, whipstock **276** may include an interior passage.

(65) Whipstock **276** may be positioned within the frac window system **226** at various different times. In at least one embodiment, the whipstock **276** may be run-in-hole with the frac window system **226**. In yet other embodiments, the whipstock **276** is run-in-hole after the frac window system **226** is run-in-hole. In yet even other embodiments, as is shown, the whipstock **276** may be run-in-hole through the high-pressure frac string **251**.

(66) Turning to FIG. 8, a straddle stimulation tool **285** is illustrated extending from the frac window system **226** into the upper secondary wellbore **12b**. In the illustrated embodiment, the straddle stimulation tool **285** is run-in-hole through the high-pressure frac string **251**. Straddle stimulation tool **285** generally includes a straddle tubular **286** having a first end and a second end forming a flow bore therebetween. Straddle tubular **286** includes an inner surface and an outer surface. When deployed, straddle stimulation tool **285** is positioned so that first end is in first wellbore **12** and second end is in secondary wellbore **12b**. In this regard, first end may be positioned within elongated tubular of frac window system **226** and second ends may be positioned within the first end of secondary wellbore casing **216**.

(67) More specifically, a first seal **292** may be disposed along the outer surface **290** adjacent the

second end. Seal **292** is disposed to engage the inner surface of secondary wellbore casing **216** to seal the annulus formed between casing **216** and straddle stimulation tool **285**. A straddle depth mechanism **294** may be disposed along the outer surface **290** of the straddle tubular **286** adjacent the first end, the straddle depth mechanism **294** engaging the first depth mechanism **248** of the frac window system **226**. A second seal **296** may be provided on the outer surface **290** of the straddle tubular **286**, the second seal **296** engaging the inner surface **234** of the elongated tubular **228** of the frac window system **226**. Second seal **296** may engage one of the smooth the sealing surfaces **254** of elongated tubular **228** to ensure an effective or desirable seal.

(68) In one or more embodiments, first seal **292** may be formed of multiple seal elements, such as first seal element spaced apart from a second seal element. A port may extend from inner surface **289** to outer surface **290** between seal elements.

(69) As shown in FIG. **9A**, the straddle stimulation tool **285** functions to isolate the portion of first wellbore **12** below window **212**, including secondary wellbore **12a**, from secondary wellbore **12b**. The seals as described permit delivery of a high-pressure fluid to upper secondary wellbore **12b** without impacting lower secondary wellbore **12a**. For example, hydraulic fracturing operations can be carried out with respect to upper secondary wellbore **12b** without impacting lower secondary wellbore **12a**. This might be desirable after one secondary wellbore **12a**, **12b** has been producing for some time and it is determined that only certain secondary wellbores within the system (such as secondary wellbore **12b**) may need stimulation, while other secondary wellbores (such as secondary wellbore **12a**) do not. In another example, since the vast majority of unconventional wellbores have to be stimulated before they are capable of producing hydrocarbons, the foregoing will allow each of wellbores **12a**, **12b** to be isolated and hydraulically fractured in order to promote production. The straddle stimulation tool **285** and the main bore isolation sleeve **260** not only isolate the wellbores **12a** and **12b** from one another, but also provide a path for balls, plugs, etc. to be dropped from the surface to isolate individual zones in the wellbores during the stimulation process.

(70) As shown in FIG. **9A**, the secondary wellbore **12b** has at least partially been stimulated. For example, the secondary wellbore **12b** has been stimulated from toe to heel, for example using one or more frac plugs **910** to isolate the different stimulated regions. In at least one embodiment, as shown, the deployment of the frac plugs **910** and the stimulation process are conducted through the high-pressure frac string **251**.

(71) Turning to FIG. **9B**, illustrated the well system of FIG. **9A** with additional features therein. For example, FIG. **9B** the SST **285** may additionally include a debris barrier **920**. The debris barrier **920**, in one embodiment, is configured to prevent proppant, fines, and/or other small items (debris) from moving (e.g., flowing into, settling into, fall into) into the area between the SST's **285** OD and the upper nipple profile of the frac window system **226**. Other areas and devices may benefit from one or more barriers, one or more types of barriers, seals, wipers, energized (self and/or pressure), etc. This is only an example of where debris may be of a concern as is mentioned only as one example.

(72) Another innovation that would play a helpful role in a high-pressure, high-flow rate, multilateral environment is sealing elements and the protection of sealing elements. In one or more embodiments, a protective sheath **930** may be utilized to protect one or more seals from sustaining damaged while being run in the well, passing from one tool/device/profile to another, etc. As for an example, a protective sheath **930** may be utilized on the lower end of the SST **285**. In this embodiment, the protective sheath **930** may be round to cover a circular seal (as an example) or other shapes or configurations. In one or more embodiments, the protective sheath **930** may slidably fit over the SST's **285** lower seals while the SST **285** is being deployed in the wellbore **12**. In some embodiment, the protective sheath **930** may consist of one or more parts. For example, the protective sheath **930** may employ a releasable device that secures it in one position/location but then releases so the sheath may move to another position (such as a position away from the seals so

that the seals may sealing engage a polished seal bore (smooth bore especially designed for the seals to engage against—and capable of withstanding the high-pressures and stresses of stimulating at pressure of above 5,000-psi (e.g., 340 atm), above 10,000-psi (e.g., 680 atm), at least 12,500-psi (e.g., 851 atm) and/or over 15,000-psi (e.g., 1021 atm). In some highly specialized situations, it may be desirable to be able to withstand pressures of 30,000-psi (e.g., 2041 atm) or more. The protective sheath **930** and/or related parts/pieces/devices such as the seal mandrel (the tubular shaped devices that holds the seals in place and allows fluid to flow the inside) may comprise one or more other securing features (such as a collet or snap ring) to secure the protective sheath **930** in a second position/location. The second position may be utilized as a position to secure the protective sheath **930** away from the seals after the seals have landed in the seal bore—this may be utilized in some, but not all embodiments. And as other features, devices, concepts disclosed, these examples are provided as examples.

(73) FIG. **9B** shows another alternative place for seals and a seal protection-device. The SST **285** in certain embodiments will have one or more seals at the upper end to seal in the frac window. One or more seal-protection devices may be used in most embodiments. In some similar or different embodiments, a seal-bore protection device may be utilized to protect the seal-bores before or during or after—or combination thereof—a high-pressure event requiring the use of the seal bore. A seal-bore protection device may also protect the sealing surfaces (etc.) from erosion, corrosion, or both. The protection device may comprise a mechanical barrier/device, a chemical barrier/coating, the use of a special CRA (Corrosion Resistant Alloy) or other material(s) or any combination thereof with the goal of protecting the seal bores from degradation during one or more phases of the drilling, completion, stimulating, production, workover of the well and/or the device(s). For example, if the frac window nipple profiles (including the sealing bores) are to be used as a service tool, it would be preferable for the seals bores to be usable for use in several wells, several stimulation jobs, and/or both. If the equipment is to be utilized as long-term production usage, the equipment may be preferable to have different characteristics than equipment designated as a service tool. For example, long-term production equipment may be made of lower strength materials such as L-80, N-80, J-55, or similar alloys. Lower-strength alloys are known to be more corrosion resistant than higher-strength alloys. On the other hand, higher-strength alloys are capable of withstanding higher pressures (12,500-psi (e.g., 851 atm), capable of withstand erosion better, etc. The above differences are presented not to limit the extent of the disclosure, but to illustrate the variability within the scope of the invention.

(74) FIG. **10** illustrates production from the upper secondary wellbore **12b** or flowback of fluids **303**, such as hydraulic fracturing fluids and/or hydrocarbons, from fractures **305** resulting from such an operation, where flowback **303** from secondary wellbore **12b** is illustrated while secondary wellbore **12a** remains isolated.

(75) FIG. **11** illustrates the placement of an upper secondary wellbore plug **1110** within the upper secondary wellbore **12b**. In the illustrated embodiment, the upper secondary wellbore plug **1110** is run-in-hole through the high-pressure frac string **251**, for example through the straddle stimulation tool **285**.

(76) FIG. **12** illustrates the removal of the straddle stimulation tool **285**. In the illustrated embodiment, the straddle stimulation tool **285** is removed through the high-pressure frac string **251**.

(77) FIG. **13** illustrates the removal of the deflector **276**. In the illustrated embodiment, the deflector **276** is removed through the high-pressure frac string **251**.

(78) FIG. **14** illustrates the removal of the plug **274**. In the illustrated embodiment, the plug **274** is removed through the high-pressure frac string **251**. In yet another embodiment, the plug **274** is drilled out through the high-pressure frac string **251**.

(79) FIG. **15A** illustrates the placement of an isolation sleeve **1560** within the frac window system **226**. The isolation sleeve **1560**, in at least one embodiment, is similar to the isolation sleeve **260**

described and illustrated with respect to FIG. 5 above. In the illustrated embodiment, the isolation sleeve **1560** is run-in-hole through the high-pressure frac string **251**.

(80) In at least one embodiment, the isolation sleeve **1560** includes an outside diameter (OD.sub.s) and an inside diameter (ID.sub.s). The outside diameter (OD.sub.s), in at least one embodiment, is at least 4.5" (e.g., 11.43 cm), if not at least 5" (e.g., 12.7 cm), if not at least 5.5" (e.g., 13.97 cm), or more. Again, the outside diameter (OD.sub.s) may be based upon the inside diameter (ID) of the frac window system **226** it is configured to engage with. In at least one embodiment, the isolation sleeve **1560** includes an inside diameter (ID.sub.s) of at least 4" (e.g., 10.16 cm), if not at least 4.5" (e.g., 11.43 cm), or more. In yet another embodiment, the isolation sleeve **1560** has a wall thickness (t.sub.s) and a length (L.sub.s).

(81) FIGS. **15B** through **15G** illustrate various different embodiments and views of a frac window system **400**, similar to the frac window system **400** illustrated and discussed with regard to FIGS. **4B** through **4G** above, having one embodiment of the isolation sleeve **1560** included therein.

(82) As shown in FIG. **16**, the isolation sleeve **1560** functions to isolate the portion of first wellbore **12** below window **212**, including secondary wellbore **12b**, from secondary wellbore **12a**. The seals as described permit delivery of a high-pressure fluid to lower secondary wellbore **12a** without impacting upper secondary wellbore **12b**. For example, hydraulic fracturing operations can be carried out with respect to lower secondary wellbore **12a** without impacting upper secondary wellbore **12b**. This might be desirable after one secondary wellbore **12a**, **12b** has been producing for some time and it is determined that only certain secondary wellbores within the system (such as secondary wellbore **12a**) may need stimulation, while other secondary wellbores (such as secondary wellbore **12b**) do not. In another example, since the vast majority of unconventional wellbores have to be stimulated before they are capable of producing hydrocarbons, the foregoing will allow each of wellbores **12a**, **12b** to be isolated and hydraulically fractured in order to promote production. The isolation sleeve **1560** not only isolates the wellbores **12a** and **12b** from one another, but also provide a path for balls, plugs, etc. to be dropped from the surface to isolate individual zones in the wellbores during the stimulation process.

(83) As shown in FIG. **16**, the lower secondary wellbore **12a** has at least partially been stimulated. For example, the lower secondary wellbore **12a** has been stimulated from toe to heel, for example using one or more frac plugs **1610** to isolate the different stimulated regions. In at least one embodiment, as shown, the deployment of the frac plugs **1610** and the stimulation process are conducted through the high-pressure frac string **251**.

(84) FIG. **17** illustrates production from the lower secondary wellbore **12a** or flowback of fluids **1703**, such as hydraulic fracturing fluids and/or hydrocarbons, from fractures **1705** resulting from such an operation, where flow from the lower secondary wellbore **12a** is illustrated while the upper secondary wellbore **12b** remains isolated.

(85) FIG. **18** illustrates the removal of the isolation sleeve **1560** and the lateral plug **1110**. In the illustrated embodiment, the isolation sleeve **1560** and the lateral plug **1110** are removed through the high-pressure frac string **251**. What results is commingled flow from the lower secondary wellbore **12a** and the upper secondary wellbore **12b**. In at least one embodiment, the commingled flow travels to the surface of the wellbore **12** through the frac window system **226** and the high-pressure frac string **251**. Accordingly, the high-pressure frac string **251** may ultimately function as production tubing, for at least as long as it remains within the wellbore **12**.

(86) FIG. **19A** illustrates an alternative embodiment of the disclosure, wherein a sleeve **1910**, including a tubular **1920** having one or more flow control orifices **1930**, is positioned within the frac window system **226**. In at least one embodiment, the sleeve **1910** includes two or more flow control orifices that are located in a sidewall of the sleeve **1910** and restrict the flow from the upper secondary wellbore **12b**. Depending on the amount of flow desired from the upper secondary wellbore **12b**, the number and/or size of the one or more flow orifices **1930** may be adjusted. If the operator decides a different-size and/or number of orifices **1930** are required, they may pull the

sleeve **1910** and replace it with one with different-sized and/or numbered orifices. The sleeve **1910**, in at least one embodiment, may be installed and/or removed and/or replaced through the high-pressure frac string **251**.

(87) FIG. **19B** illustrates a zoomed in view of the sleeve **1910**. As shown, the sleeve **1910** includes a tubular **1920** having a first tubular end **1920a** and a second tubular end **1920b**. In at least one embodiment, the tubular **1920** has an outside diameter (OD.sub.s) of at least 5.5". As shown, the isolation sleeve includes the one or more flow control orifices **1930** (e.g., two or more flow control orifices **1930**) located in a sidewall of the tubular **1920** between the first tubular end **1920a** and the second tubular end **1920b**. Accordingly, as shown, the tubular **1920** is configured to be placed within the frac window system **226** at a junction between a first wellbore **12** and a secondary wellbore **12b** such that the flow control orifices **1930** restrict a flow of wellbore fluid from the secondary wellbore **12b** into the elongated tubular.

(88) In the illustrated embodiment, the sleeve **1910** further includes an uphole seal mandrel **1940** including a first uphole seal **1940a** and a second uphole seal **1940b** located at least partially along an outer surface of the tubular **1920** proximate the first tubular end **1920a**, and a downhole seal mandrel **1950** including a first downhole seal **1950a** and second downhole seal **1950b** located at least partially along the outer surface of the tubular **1920** proximate the second tubular end **1920b**.

(89) As shown, the sleeve **1910** may further include a running/retrieving mechanism **1960**, as well as a depth mechanism **1965**, as well as a locking profile **1970**. The running/retrieving mechanism **1960** is designed so that a running and/or retrieving tool may be releasably attached to the sleeve **1910**. The running tool may be coupled to sleeve **1910** so that sleeve **1910** can be deployed in the well and landed in the depth mechanism **1965** disposed along the inner surface **234** of elongated tubular **228** of frac window system **226** (FIG. **19A**). The depth mechanism **1965** (e.g., **272** in **19A**) may be positioned between the first uphole seal **1940a** and the first tubular end **1920a**.

(90) In one or more embodiment, the one or more orifices **1930**, may be disposed in one or more tubulars **1920**. FIG. **19B** illustrates only one tubular **1920**, but multiple tubulars may be utilized. Since tubulars **1920** may see considerable erosion, it may be more cost-efficient to change out only the most-eroded tubulars **1920** in lieu of changing out one long tubular member **1920**.

(91) In one or more embodiment, orifices **1930** may comprise tungsten, a tungsten carbide, ceramic, one or more carbides, and/or one or more other erosion-resistant elements or compounds thereof. In one or more embodiment, orifices **1930** and related members may comprise straight flow paths, circular flow paths, holes, shaped inlets, shaped outlets, or combinations thereof. In one or more embodiments, orifices **1930** may be secured via an interference fit (e.g., press fit, driving fit, forced fit) with the one or more tubulars **1920**. Other methods may be utilized separately, or in concert, including a chemical bond using such materials as thread-locking formulas which may be methacrylate-based and rely on the electrochemical activity of a metal substrate to cause polymerization of the fluid; Brazing, and Welding, etc. Brazing joins two metals by heating and melting a filler (alloy) that bonds to the two pieces of metal and joins them. Welding uses high temperatures to melt and join two metal parts.

(92) In one or more embodiments, orifices **1930** may be oriented in a preferred direction so that the orifices **1930** will be aligned with opening **230** (shown in FIG. **5**) which is aligned with wellbore **12b** as shown in FIGS. **5** and **19A**. When it is preferred to align orifices **1930** with opening **230**, then a means to align the sleeve **1910** with opening **230** may be desirable. In one or more embodiments, sleeve **1910** may have one or more devices for the alignment. For example, an orientation device like **4052** used in **4058** (FIG. **40D**) may be utilized. Another example is **4094** as shown in FIG. **40F** may be implemented. In other embodiments, the orientation may happen by means of other devices related to the running tool, Frac Window **26**, etc. Orifices **1930** maybe of any size or shape, mounted in a wall of the sleeve **1910** or integral to one or more tubulars **1920**.

(93) In one or more embodiments, the window exit **420** may have grooves or slots formed on the OD or the ID surface so flow may easily enter orifices **1930** even if they are not aligned with

opening **230** (shown in FIG. 5). In other embodiments, the window exit **420** may have holes extending from the OD to the ID to ensure flow may enter orifices **1930** without impediment. In some embodiments, the holes extending from the OD to the ID may intersect grooves or slots formed on the OD of **420**, grooves or slots formed on the ID of **420**, or both grooves or slots formed on the OD and the ID surfaces of window exit **420**. The grooves or slots may be of any one or more orientation (circumferential, longitudinal, angular, etc.)

(94) In one or more embodiments, the holes extending from the OD to the ID of **420** may comprise devices to control flow, regulate flow, modify flow, impede flow, filter the flow, separate the flow into one or more constituents, measure one or more parameters of the flow (temperature, solids content, water content, pressure, change in pressure, density, velocity, radiation, conductivity, refractance, reflectance, etc.).

(95) In one or more embodiments, frac window system **400** or one or more of its components (e.g., **420**) may comprise devices, to control flow, regulate flow, modify flow, impede flow, filter the flow, separate the flow into one or more constituents, measure one or more parameters of the flow (temperature, solids content, water content, pressure, change in pressure, density, velocity, radiation, conductivity, wave reflectance, reflectance, etc.).

(96) It may be desirable to control the production from lateral **12b** and lateral **12a** (and/or other wellbores located below **12b**). In FIGS. **19C** and **19D**, an orifice insert **1975** is deployed inside of sleeve **1910**. In some embodiments, orifice insert **1975** may be located proximate the second tubular end **1920b** (lower end) of sleeve **1910** so that the flow from a lower wellbore (e.g., **12a**) may be restricted and/or controlled (independently from **12b**).

(97) In some embodiments, the orifice insert **1975** may be fixedly-releasable from sleeve **1910**. In some embodiments, the orifice insert **1975** may be mounted to a holder **1980** that can provide a means to retrieve and replace the orifice insert **1975** without pulling the sleeve **1910**. In some embodiments, the holder **1980** may be releasably-fixed to sleeve **1910** by locking profile **1970** shown in FIG. **19C**.

(98) As shown in **19D**, one or more seals **1982** may be used with the orifice insert **1975** and holder **1980**. Furthermore, a retaining device **1984** may be used to retain the orifice insert **1975** within the holder **1980**.

(99) In some embodiments it may be desirable to change all orifices—including the orifice insert **1975** and the orifices **1930** controlling flow from lateral **12b**—without pulling the entire sleeve **1910**. In FIGS. **19E** and **19F**, the holder **1980** has been exchanged for a long holder **1980b**. In this embodiment, the long holder **1980b** may include the orifice insert **1975** as well as replacement orifices **1930b** placed adjacent to secondary wellbore **12b** (and opening **230** of **26**).

(100) FIGS. **19E** and **19F** illustrate the replacement orifices **1930b** aligned with orifices **1930** of the sleeve **1910**. The replacement orifices **1930b** may be aligned by one or methods or devices, such as for example a lug or similar device (e.g., which may be similar follower **281** illustrated in FIG. **7**). The lug or similar device (not shown) may be positioned along the outer surface of the tool used to deploy it into the well. The lug or similar device may protrude from an outer surface to engage orientation device in order to rotate the replacement orifices **1930b** to the desired angular position within sleeve **1910**.

(101) Other ways may be utilized to orient the replacement orifices **1930b**. In some embodiments the use of a running tool encompassing an electronic orientation device (e.g. gyroscope, 6-axis or 9-axis Inertial Measurement Unit (IMU), Measurement While Drilling (MWD) directional instruments, Halliburton's Work String Orientation Tool (WOT), etc.) may be used. Likewise, the running tool may also encompass a rotational device to physically rotate the replacement orifices **1930b** to the desired orientation. The rotational device may be used with or without the aid of an electronic, mechanical, or other orientation devices.

(102) In some embodiments, the orifices **1930**, **1930b** and **1975** are passive components. These orifices **1930**, **1930b** and **1975** may employ a specific setting to partially choke flow. The resulting

arrangement can be used to delay water or gas breakthrough by reducing production from a selected wellbore (e.g., **12a**, **12b**, etc.).

(103) In some embodiments, the sleeve **1910** comprises an actuatable component **1985** (e.g., valve) installed as part of a well completion to help optimize production, for example, by actively equalizing reservoir inflow from one or more wellbores (e.g., **12a**, **12b**, etc.). An actuatable component **1985** may be a spring-type component that adjusts the orifice sized as a function of the pressure differential.

(104) Other actuatable components may comprise remotely-actuated devices (e.g., to change the size/area of the flow area and/or orifice) (e.g., remote flow control valve, interval control valve, etc.) or other parameter/mechanism. The active components may be actuated via pressure pulses, time intervals, temperature changes, fluid changes, etc. (or combination thereof) from the surface or other location.

(105) In some other embodiments, the orifices **1930**, **1930b** and **1975** may comprise embodiments of an autonomous control device similar to Halliburton's EquiFlow® Autonomous In Flow Control Device (AICD). The orifices **1930**, **1930b** and **1975** may comprise an entry shape and/or internal design to create a rotational flow. Oil may exit the orifices with a pressure drop very similar to that of a passive (e.g., fixed flow area) orifice while gas and water rotate at a higher velocity because of their lower viscosities, creating low pressure in the core region that causes flow breakdown. As a result, the flow rates of gas and water are reduced. Another benefit of the higher velocity of gas and water is that the effect increases as the amount of gas and water in the flow stream increases.

(106) FIG. **19I** illustrates the area (or volume) that is available for implementing an autonomous feature. In some embodiments, the area/volume may be used to implement actuatable components. By utilizing one or more control devices in wellbore **12a**, **12b**, and other connected wellbores the flow of oil, gas, water and combinations thereof may be regulated and controlled. When used in concert with sleeve **1910**, and **226**, the production of oil, gas and water can be optimized to maximize hydrocarbon recovery and increase economic returns during various stages of the multilateral well's life (e.g., natural flow, during secondary recovery operations (e.g., using ESP's, Rod Pumps, Gas Lift, etc.) and during tertiary recovery and/or disposal operations (e.g., water flood, CO₂ flood, chemical flood, water disposal, etc.).

(107) At some point in time, the pressures within the lower secondary wellbore **12a** and/or upper secondary wellbore **12b** may reduce to a value wherein the high-pressure frac string **251** and/or frac window system **226** may be removed from the wellbore **12**, and optionally, a less expensive conventional frac window system and/or low-pressure frac string and/or production tubing may be insert within the wellbore **12** for collecting the commingled flow from the lower secondary wellbore **12a** and the upper secondary wellbore **12b**. For example, the frac window system **226** may be left in the wellbore **12** until a later time, such as when the bottom hole pressure (BHP) is low enough that 5,000-psi (e.g., 340 atm) completion equipment may be installed. At that time, the operator would purchase 5,000-psi (e.g., 340 atm) rated equipment that is made of low-alloy steel, which has a low enough yield strength that corrosion is not a concern (e.g., L-80 material, K-55 material, etc.), and install it within the wellbore **12**.

(108) FIG. **20** illustrates the removal of the frac window system **226** and the installation of a conventional frac window system **2026** (e.g., 5,000-psi rated equipment). FIG. **20** additionally illustrates a low-pressure frac string **2051** engaging with the conventional frac window system **2026**. In an alternative embodiment, conventional production tubing, as opposed to the low-pressure frac string **2051**, may engage with the conventional frac window system **2026**.

(109) In other embodiments, a dual-string completion system, such as Halliburton's FloRite® System, which allows separate production from lower secondary wellbore **12a** and the upper secondary wellbore **12b**, may also be utilized. One advantage of the FloRite® System with Vector Block is that it allows re-entry into either lateral wellbore **12a**, **12b**. It also provides a high-pressure Level 5 hydraulic seal between the lateral wellbore **12b** and the mainbore **12**, which ensures a less

drawdown at the heel of lateral wellbore **12b**.

(110) FIGS. **21** and **22** illustrate the well system of FIG. **20** after different types of artificial lift equipment **2110**, **2210**, respectively, have been installed within the wellbore **12**. In at least one embodiment, the artificial lift equipment **2110**, **2210** is installed at the same time as the conventional frac window system **2026** and/or low-pressure frac string **2051**, thereby saving time and money.

(111) FIG. **23** illustrates the well system of FIG. **20** after the placement of an upper secondary wellbore plug **2310** within the upper secondary wellbore **12b**. In the illustrated embodiment, the upper secondary wellbore plug **2310** is run-in-hole through the low-pressure frac string **2051**. In at least one embodiment, the upper secondary wellbore plug **2310** is a high-expansion plug, as might be required to traverse the low-pressure frac string **2051** while still having the ability to seal the upper secondary wellbore **12b**.

(112) FIG. **24A** (e.g., with reference to FIG. **24B**) illustrates the placement of an isolation sleeve **2460** within the convention frac window system **2026**. The isolation sleeve **2460**, in at least one embodiment, is similar in many respects to the isolation sleeve **260** described and illustrated with respect to FIG. **5** above. For example, the isolation sleeve **2460** includes a tubular **2465** having a first tubular end **2465a** and a second tubular end **2465b**. However, the isolation sleeve **2460** additionally includes a first high-expansion seal **2470** located at least partially along an outer surface of the tubular **2465** proximate the first tubular end **2465a** and a second high-expansion seal **2475** located at least partially along the outer surface of the tubular **2465** proximate the second tubular end **2465b**. The first and second high-expansion seals **2470**, **2475**, in the illustrated embodiment, are configured to move between a radially retracted state (FIGS. **24A-24C**) for running the isolation sleeve **2460** in hole and a radially expanded state (FIGS. **25A-25C**) for engaging an inner surface of the frac window system **2026**. Accordingly, the isolation sleeve **2460** may be similar to the isolation sleeve **260**, but for the inclusion of the high-expansion seals **2470**, **2475** and the smaller outside diameter necessary to traverse the low-pressure frac string **2051**. In the illustrated embodiment, the first high-expansion seal **2470** and the second high-expansion seal **2475** are spaced such that they isolate (e.g., span) a junction between the first wellbore **12** and the secondary wellbore **12b**. In the illustrated embodiment, the isolation sleeve **2460** is run-in-hole through the low-pressure frac string **2051**. In the embodiment of FIG. **24A**, the high-expansion seals **2470**, **2475** are in their radially retracted state.

(113) FIG. **24B** illustrates an enlarged cross-sectional view of the isolation sleeve **2460** of FIG. **24A** having the high-expansion seals **2470**, **2475**. In the illustrated embodiment, the isolation sleeve **2460** further includes a depth mechanism **2480** disposed at least partially along the outer surface of the tubular **2465**, the depth mechanism **2480** configured to engage a related depth mechanism disposed along an inner surface of the frac window system **2026**. In at least one embodiment, the depth mechanism **2480** is positioned between the first high-expansion seal **2470** and the first tubular end **2465a**.

(114) In the illustrated embodiment of FIG. **24B**, the isolation sleeve **2460** includes an inside diameter (ID.sub.6) and an outside diameter (OD.sub.6). The outside diameter (OD.sub.6), in at least one embodiment, is at least 0.95" (e.g., 2.413 cm), if not at least 2.867" (e.g., 7.28 cm), if not at least 4.545" (e.g., 11.54 cm), if not at least 6.151" (e.g., 15.62 cm), if not at least 12.313" (e.g., 31.28 cm), or more. Again, the outside diameter (OD.sub.6) may be based at least in part upon the inside diameter of the low-pressure frac string **2051** that the isolation sleeve **2460** must traverse. The inside diameter (ID.sub.6), in at least one embodiment, is at least 0.47" (e.g., 1.19 cm), if not at least 2.398" (e.g., 6.09 cm), if not at least 3.96" (e.g., 10.06 cm), if not at least 5.36" (e.g., 13.61 cm), if not at least 11.081" (e.g., 28.15 cm), or more. In at least one embodiment, the isolation sleeve **2460** comprises at least 55-ksi grade material that can accommodate an internal yield pressure of at least 5,000-psi (e.g., 340 atm).

(115) In yet another embodiment, the isolation sleeve **2460** has a wall thickness (t.sub.6). In at least

one embodiment, the wall thickness (t.sub.6) is at least 0.2" (e.g., 0.51 cm), if not at least 0.7" (e.g., 1.78 cm), if not at least 2.0" (e.g., 5.08 cm). In at least one embodiment, the wall thickness (t.sub.6) ranges from 0.22" (e.g., 0.56 cm) to 0.634" (e.g., 1.61 cm).

(116) The term high-expansion, as used herein, means that the feature expands by at least 3% when moving from its radially reduced state to its radially expanded state. In yet another embodiment, the term extremely high-expansion, as used herein, means that the feature expands by at least 10% when moving from its radially reduced state to its radially expanded state. In yet another embodiment, the term super high-expansion, as used herein, means that the feature expands by at least 15%, if not at least 20%, or not at least 25%, when moving from its radially reduced state to its radially expanded state. Unless otherwise stated, any reference to high-expansion in this document may encompass all of the above percentages.

(117) FIG. 24C illustrates an isometric view of an alternative embodiment of the isolation sleeve 2460 of FIG. 24A having the high-expansion seals 2470, 2475 in their radially retracted state.

(118) FIG. 25A illustrates the expansion of the high-expansion seals 2470, 2475 of the isolation sleeve 2460 within the convention frac window system 2026. Accordingly, as shown, the high-expansion seals 2470, 2475 are in their radially expanded state.

(119) FIG. 25B illustrates an enlarged cross-sectional view of the isolation sleeve 2460 of FIG. 25A having the high-expansion seals 2470, 2475 in their radially expanded state.

(120) FIG. 25C illustrates an isometric view of an alternative embodiment of the isolation sleeve 2460 of FIG. 25A having the high-expansion seals 2470, 2475 in their radially expanded state.

(121) FIG. 25D, with reference to 24D, illustrates a cross-sectional view of an alternative embodiment of the isolation sleeve of FIG. 24D having the high-expansion seals in their radially expanded state (FIG. 25D) and radially retracted state (FIG. 24D). In at least one embodiment, the high-expansion seals 2510 may require the compression of the elastomer seals so that the outside diameter (D.sub.s) will increase as the length (L.sub.s) of the elastomer seals decrease. In one or more embodiment this may require a 2-part mandrel so one end of the elastomer seals remains fixed while the other end is compressed by the movement of a 2-part mandrel. The dynamic portion 2520 of the 2-part mandrel is in the extended position when the assemble is being run into the well. When a specific load is applied to the dynamic portion, it is released via one or more shear devices (not shown) As the dynamic portion 2520 travels downwards, the high-expansion seals 2510 are compressed axially; and the length (L.sub.s) of the elastomer decreases. In most embodiments, the high-expansion seals 2510 are nearly incompressible meaning that the volume of the high-expansion seals does not change under load. Consequently, as length (L.sub.s) of the elastomer decreases, the outside diameter (D.sub.s) increases. In some embodiments, the gap 2560 between the OD of the isolation sleeve and the ID of the item which the elastomer will seal against requires a bridging device at each end of the elastomer seals. Without a bridging device, the elastomer seals may extrude (flow) away from the ID which it is intended to seal against. This extruding of the elastomer may result in inadequate pressure between the ID it is to seal against and the elastomer itself. This pressure is often referred to as the sealing pressure. If the sealing pressure is less than the pressure of the fluid that is trying to pass by the elastomer seal, the seal is inclined to fail. Therefore, in a bridging device is typically needed for the elastomer seals to affect enough sealing pressure. In one or more embodiments, the bridging device may be an articulating bridging device like 2570 that increases in OD as it is compressed axially.

(122) Once the high-expansion seals 2510 are forced outwards and pressed against the seal surface 2550, a locking mechanism may be activated to lock dynamic portion 2520 and static portion 2530 of the 2-part mandrel together. In one or more embodiments, the locking mechanism may comprise an internal slip 2580 which has a toothed-profile which will lock into a mating toothed profile 2582. In some embodiments, the locking mechanism may comprise a releasing feature to unlock the 2-part mandrel so the high-expansion seals 2510 may relax and the assembly retrieved from the well. As shown, one or more seals 2575 may be utilized to provide a pressure barrier between the

dynamic portion **2520** and static portion **2530** of the 2-part mandrel. The things shown and discussed here are only examples of things that can be utilized to compress, lock a force/load into the high-expansion seals **2510** (referred to as energizing the seals), seal the path between the 2-part mandrel, etc. In other embodiments, the articulating bridging device may be replaced with “petal plates”. Petal plates are a set of more than one set of plates that have the shape of petals of a flower. Just like the petals of a flower can expand outwards rather easily, the petal plates expand outwardly while the high-expansion seals **2510** are being compressed. The petal plates, in some embodiments, expand outwardly enough to contact the seal surface **2550**. In some embodiments, the edges of the petal plate that contact the seal surface **2550** maybe supported by portions of the high-expansion seals **2510** creating a high-pressure seal with the petal plate edges trapped between the portions of the high-expansion seals **2510** and the seal surface **2550**. Since the petal plate is trapped, the extrusion gap is zero, which means the seal material cannot extrude between the petal plate and seal surface **2550**. In one or more embodiments, the tool, or running tool, may have centralizing device that will force the tool into a centralized location within the body it is to seal against. One of the reasons for a centralizing device is to ensure the seals are not laying lowside in the well. Laying lowside would require unequal radial load to the seals since one side of the seals would be required to lift the tool and simultaneously effect a seal. By centralizing the tool prior to “setting” the seals, the seals are loading evenly around the circumference and a uniform seal pressure is created.

(123) As shown in FIG. **26**, the isolation sleeve **2460** functions to isolate the portion of first wellbore **12** below window **212**, including secondary wellbore **12b**, from secondary wellbore **12a**. The seals, as described above, permit delivery of a high-pressure fluid to lower secondary wellbore **12a** without impacting upper secondary wellbore **12b**. For example, hydraulic fracturing operations can be carried out with respect to lower secondary wellbore **12a** without impacting upper secondary wellbore **12b**. This might be desirable after one secondary wellbore **12a**, **12b** has been producing for some time and it is determined that only certain secondary wellbores within the system (such as secondary wellbore **12a**) may need stimulation, while other secondary wellbores (such as secondary wellbore **12b**) do not. The isolation sleeve **2460** not only isolates the wellbores **12a** and **12b** from one another, but also provide a path for balls, plugs, etc. to be dropped from the surface to isolate individual zones in the wellbores during the stimulation process.

(124) As shown in FIG. **26**, the lower secondary wellbore **12a** has at least partially been re-stimulated. For example, the lower secondary wellbore **12a** has been re-stimulated from toe to heel, for example using one or more frac plugs **2610** to isolate the different stimulated regions. In at least one embodiment, as shown, the deployment of the frac plugs **2610** and the re-stimulation process are conducted through the low-pressure frac string **2051**.

(125) FIG. **27** illustrates production from the lower secondary wellbore **12a** or flowback of fluids **2703**, such as hydraulic fracturing fluids and/or hydrocarbons, from fractures **2705** resulting from such an operation, where flowback **2703** from the lower secondary wellbore **12a** is illustrated while the upper secondary wellbore **12b** remains isolated.

(126) FIG. **28** illustrates the removal of the isolation sleeve **2460**. In the illustrated embodiment, the isolation sleeve **2460** is removed through the low-pressure frac string **2051**. For example, the high-expansion seals **2470**, **2475** may be returned from their radially expanded state to their radially retracted state, such that they have an outside diameter (OD) small enough that the isolation sleeve **2460** may be removed through the low-pressure frac string **2051**.

(127) FIG. **29** illustrates the placement of a plug **2974** in the lower secondary wellbore **12a**. Much in the same way that isolation sleeve **2460** is utilized to isolate secondary wellbore **12b**, the plug **2974** may be deployed to isolate secondary wellbore **12a** from pumping operations relating to secondary wellbore **12b**. In at least one embodiment the plug **2974** may be positioned within frac window system **2026**, preferably at a location adjacent end **228b** or may be positioned in casing **206** of secondary wellbore **12a** or within PBR **215** (FIG. **5**) if present.

(128) FIG. **30A** illustrates the placement of a whipstock **3076** having one or more high-expansion

members **3085** in the frac window system **2026**. The whipstock **2076** may be similar in many respects to the whipstock **276** of FIG. 7 above, but for the whipstock **2076** having the one or more high-expansion members **3085** and the smaller outside diameter (OD.sub.6) necessary to traverse the low-pressure frac string **2051**. The whipstock **3076**, in the illustrated embodiment, is run-in-hole through the low-pressure frac string **2051**. In the embodiment of FIG. 30A, the one or more high-expansion members **3085** are in their radially retracted state.

(129) FIG. 30B illustrates an enlarged cross-sectional view of the whipstock **3076** of FIG. 30A having high-expansion members **3085**. As shown, the whipstock **3076** may include a housing **3080** having a first whipstock end **3080a** with a contoured surface **3082** and a second whipstock end **3080b**. As further shown, the whipstock **3076** may include the one or more high-expansion members **3085** located at least partially along an outer surface of the housing **3080**. As discussed, the one or more high-expansion members **3085** are configured to move between a radially retracted state (FIGS. 30A and 30B) for running the whipstock **3076** in hole and a radially expanded state (FIGS. 31A and 31B) for engaging an inner surface of a frac window system **2026**.

(130) In the illustrated embodiment, the one or more high-expansion members **3085** are one or more scissor type high-expansion members. Nevertheless, other types of high-expansion members are within the scope of the disclosure. As show, the one or more high-expansion members **3076** may each have a plurality of teeth **3090** for engaging the inner surface of the frac window system **2026**.

(131) The whipstock **3076** is illustrated as a neckless whipstock. However, the whipstock **3076** could be configured as a necked whipstock or extended necked whipstock and remain within the scope of the present disclosure. As further shown, the whipstock **3076** may further include a depth mechanism **3092** disposed at least partially along the outer surface of the housing **3080**, the depth mechanism **3092** configured to engage a related depth mechanism disposed along an inner surface of the frac window system **2026**. In at least one embodiment, the depth mechanism **3092** is positioned between the one or more high-expansion members **3076** and the second whipstock end **3080b**.

(132) In the illustrated embodiment of FIG. 31B, the whipstock **3076** includes outside diameter (OD.sub.6). The outside diameter (OD.sub.6), in at least one embodiment, is at least 0.95" (e.g., 2.413 cm), if not at least 2.992" (e.g., 7.60 cm), if not at least 4.583" (e.g., 11.64 cm), if not at least 8.379" (e.g., 21.28 cm), if not at least 12.313" (e.g., 31.28 cm), or more. Again, the outside diameter (OD.sub.6) may be based upon the inside diameter of the low-pressure frac string **2051** that the isolation sleeve **2460** must traverse.

(133) FIG. 31A illustrates the expansion of the high-expansion members **3085** of the whipstock **2076** within the convention frac window system **2026**. Accordingly, as shown, the high-expansion members **3085** are in their radially expanded state.

(134) FIG. 31B illustrates an enlarged cross-sectional view of the whipstock **2076** of FIG. 32A having the high-expansion members **3085** in their radially expanded state.

(135) Turning to FIG. 32, a straddle stimulation tool **3285** is illustrated extending from the frac window system **2026** into the upper secondary wellbore **12b**. In the illustrated embodiment, the straddle stimulation tool **3285** is run-in-hole through the low-pressure frac string **2051**. Straddle stimulation tool **3285** is similar in many respects to the straddle stimulation tool **285**, but for the straddle stimulation tool **3285** may include high-expansion seals **3290**. The high-expansion seals **3290**, in the illustrated embodiment, are in their radially retracted state, and may be similar in one fashion or another to the high-expansion seals **2465** of the isolation sleeve **2460**.

(136) FIG. 33 illustrates the expansion of the high-expansion seals **3290** of the straddle stimulation tool **3285**. Accordingly, as shown, the high-expansion seals **3290** are in their radially expanded state.

(137) FIG. 34 illustrates the removal of the plug **2310**. Those skilled in the art understand the processes that might be used to remove the plug **2310**.

(138) As shown in FIG. 35, the secondary wellbore **12b** has at least partially been re-stimulated. For example, the secondary wellbore **12b** has been re-stimulated from toe to heel, for example using one or more frac plugs **3510** to isolate the different stimulated regions. In at least one embodiment, as shown, the deployment of the frac plugs **3510** and the re-stimulation process are conducted through the low-pressure frac string **2051**.

(139) FIG. 36 illustrates production from the upper secondary wellbore **12b** or flowback of fluids **3603**, such as hydraulic fracturing fluids and/or hydrocarbons, from fractures **3605** resulting from such an operation, where flowback **3603** from secondary wellbore **12b** is illustrated while secondary wellbore **12a** remains isolated.

(140) FIG. 37 illustrates the removal of the straddle stimulation tool **3285**. In the illustrated embodiment, the straddle stimulation tool **3285** is removed through the low-pressure frac string **2051**. For example, the high-expansion seals of the straddle stimulation tool **3285** may be returned from their radially expanded state to their radially retracted state, such that they have an outside diameter (OD) small enough that the straddle stimulation tool **3285** may be removed through the low-pressure frac string **2051**.

(141) FIG. 38 illustrates the removal of the deflector **3076**. In the illustrated embodiment, the deflector **3076** is removed through the low-pressure frac string **2051**. For example, the high-expansion members **3085** may be returned from their radially expanded state to their radially retracted state, such that they have an outside diameter (OD) small enough that the deflector **3076** may be removed through the low-pressure frac string **2051**.

(142) FIG. 39 illustrates the removal of the plug **2974**. In the illustrated embodiment, the plug **2974** is removed through the low-pressure frac string **2051**. In yet another embodiment, the plug **2974** is drilled out through the low-pressure frac string **2051**. What results is commingled flow from the lower secondary wellbore **12a** and the upper secondary wellbore **12b**. In at least one embodiment, the commingled flow travels to the surface of the wellbore **12** through the frac window system **2026** and the low-pressure frac string **2051**. Accordingly, the low-pressure frac string **2051** may ultimately function as production tubing, for at least as long as it remains within the wellbore **12**.

(143) In certain situations, the pressures within the lower secondary wellbore **12a** and/or upper secondary wellbore **12b** may reduce to a value wherein the high-pressure frac string **251** may be removed from the wellbore **12**, and optionally, a less expensive low-pressure frac string is deployed, for example while continuing to use the frac window system **226**. As the frac window system **226** has a larger inside diameter (ID), it may become desirable and/or necessary to insert a spacer window sleeve within the frac window system **226** (e.g., prior to pulling the high-pressure frac string **251**), such that smaller outside diameter (OD) wellbore tools may traverse through the smaller inside diameter (ID) low-pressure frac string but still engage the larger inside diameter (ID) of the frac window system. The spacer window sleeve may also maintain the full function of an IsoRite® System with the use of standard (e.g., already developed) IsoRite® Tools, even though the larger frac window system **226** of the present disclosure is being used.

(144) FIG. 40A illustrates the well system of FIG. 18 after having installed a spacer window sleeve **4010** within the frac window system **226**. In the illustrated embodiment, the spacer window sleeve **4010** has been installed through the high-pressure frac string **251**.

(145) Turning to FIG. 40B, illustrated is an enlarged cross-sectional view of the spacer window sleeve **4010** of FIG. 40A. The spacer window sleeve **4010**, in at least one embodiment, includes a tubular **4020** having a first tubular end **4020a** and a second tubular end **4020b**. In accordance with one embodiment of the disclosure, the spacer window sleeve **4010** additionally includes a second opening **4030** defined in a second wall of the tubular **4020** between the first tubular end **4020a** and the second tubular end **4020b**. The second wall, in the illustrated embodiment, has a second inner surface and a second outer surface. The second opening **4030**, in at least one embodiment, aligns with the window exit **420** in the frac window system **226**.

(146) In at least one embodiment, the spacer window sleeve **4010** includes an outside diameter

(OD.sub.7) and an inside diameter (ID.sub.7). The outside diameter (OD.sub.7), in at least one embodiment, is at least 0.73" (e.g., 1.85 cm), if not at least 0.95" (e.g., 2.41 cm), if not at least 4.25" (e.g., 10.8 cm), if not at least 5.76" (e.g., 14.63 cm), if not at least 12.19" (e.g., 30.96 cm), or more. Again, the outside diameter (OD.sub.7) may be based upon the inside diameter (ID.sub.3) of the frac window system **226** it is configured to engage with. In at least one embodiment, the spacer window sleeve **4010** includes an inside diameter (ID.sub.7) of at least 0.5" (e.g., 1.27 cm), if not at least 1.9" (e.g., 4.83 cm), if not at least 2.867" (e.g., 7.28 cm), if not at least 4.5" (e.g., 11.43 cm), if not at least 10.93" (e.g., 27.76 cm), or more. In yet another embodiment, the spacer window sleeve **4010** has a wall thickness (t.sub.7) and a length (L.sub.7).

(147) Turning to FIGS. **40C** and **40D**, illustrated are an external view and cross-sectional view, respectively, of one embodiment of a spacer window sleeve **4050** with top orientation device. The spacer window sleeve **4050** includes, without limitation: an orientation device **4052**—e.g., muleshoe; anchoring device **4054**—collet; seal mandrel with external seals **4056**—5.763" (e.g., 14.64 cm); internal R-nipple profile w/ 4.525" (e.g., 11.49 cm) polish bore (may be integral with seal mandrel **4056** as shown); tubing window **4058**—window opening to pass tools out into the lateral; internal profile for landing whipstock—4.525" (e.g., 11.49 cm) (not shown, but may be integral with lower seal mandrel as shown); internal polish bore **4060**—4.525" (e.g., 11.49 cm); and seal mandrel with external seals **4062**—5.763" (e.g., 14.64 cm). In the illustrated embodiment, the orientation device **4052** is located proximate the first tubular end **4020a**.

(148) Turning to FIGS. **40E** and **40F**, illustrated are an external view and cross-sectional view, respectively, of an alternative embodiment of a spacer window sleeve **4090** with lower orientation device. The spacer window sleeve **4090** includes, without limitation: anchoring device **4091**—collet; seal mandrel with external seals **4092**—5.763" (e.g., 14.64 cm); 3) internal R-nipple profile w/ 4.525" (e.g., 11.49 cm) polish bore (may be integral with seal mandrel **4091** as shown); tubing window **4093**—window opening to pass tools out into the lateral; internal profile for landing whipstock—4.525" (e.g., 11.49 cm) (not shown); orientation device **4094**—e.g., muleshoe; internal polish bore **4095**—4.525" (e.g., 11.49 cm); and 8) seal mandrel with external seals **4096**—5.763" (e.g., 14.64 cm). In the illustrated embodiment, the orientation device **4094** is located between the second opening **4093** and the second tubular end **4020b**.

(149) Turning to FIGS. **40G** through **40M**, illustrated are various different views of the installed spacer window sleeve **4050**.

(150) FIG. **41** illustrates the removal of the high-pressure frac string **251**.

(151) FIG. **42** illustrates the installation of a low-pressure frac string **4251**, or in certain embodiments, production tubing. The low-pressure frac string **4251**, in the illustrated embodiment, engages with the PBR **255** of the frac window system **226**.

(152) FIG. **43** illustrates the well system of FIG. **42** after the placement of an upper secondary wellbore plug **4310** within the upper secondary wellbore **12b**. In the illustrated embodiment, the upper secondary wellbore plug **4310** is run-in-hole through the low-pressure frac string **4251**. In at least one embodiment, the upper secondary wellbore plug **4310** is a high-expansion plug, as might be required to traverse the low-pressure frac string **4251** while still having the ability to seal the upper secondary wellbore **12b**.

(153) FIG. **44** illustrates the placement of an isolation sleeve **4460** within the frac window system **226**. The isolation sleeve **4460**, in at least one embodiment, is similar in many respects to the isolation sleeve **2460** described and illustrated with respect to FIG. **24A** above, but for the exclusion of the high-expansion seals **2465** (e.g., the high-expansion seals are not necessary because of the spacer window sleeve **4010**). In the illustrated embodiment, the isolation sleeve **4460** is run-in-hole through the low-pressure frac string **4251**.

(154) As shown in FIG. **45**, the isolation sleeve **4460** functions to isolate the portion of first wellbore **12** below window **212**, including secondary wellbore **12b**, from secondary wellbore **12a**. The seals, as described above, permit delivery of a high-pressure fluid to lower secondary wellbore

12a without impacting upper secondary wellbore **12b**. For example, hydraulic fracturing operations can be carried out with respect to lower secondary wellbore **12a** without impacting upper secondary wellbore **12b**. This might be desirable after one secondary wellbore **12a**, **12b** has been producing for some time and it is determined that only certain secondary wellbores within the system (such as secondary wellbore **12a**) may need stimulation, while other secondary wellbores (such as secondary wellbore **12b**) do not. The isolation sleeve **4460** not only isolates the wellbores **12a** and **12b** from one another, but also provide a path for balls, plugs, etc. to be dropped from the surface to isolate individual zones in the wellbores during the stimulation process.

(155) As shown in FIG. **45**, the lower secondary wellbore **12a** has at least partially been re-stimulated. For example, the lower secondary wellbore **12a** has been re-stimulated from toe to heel, for example using one or more frac plugs **4510** to isolate the different stimulated regions. In at least one embodiment, as shown, the deployment of the frac plugs **4510** and the re-stimulation process are conducted through the low-pressure frac string **4251**.

(156) FIG. **46** illustrates production from the lower secondary wellbore **12a** or flowback of fluids **4603**, such as hydraulic fracturing fluids and/or hydrocarbons, from fractures **4605** resulting from such an operation, where flowback **4603** from the lower secondary wellbore **12a** is illustrated while the upper secondary wellbore **12b** remains isolated.

(157) FIG. **47** illustrates the removal of the isolation sleeve **4460**. In the illustrated embodiment, the isolation sleeve **4460** is removed through the low-pressure frac string **4251**.

(158) FIG. **48** illustrates the placement of a plug **4874** in the lower secondary wellbore **12a**. Much in the same way that isolation sleeve **4460** is utilized to isolate secondary wellbore **12b**, the plug **4874** may be deployed to isolate secondary wellbore **12a** from pumping operations relating to secondary wellbore **12b**. In at least one embodiment the plug **4874** may be positioned within frac window system **226**, preferably at a location adjacent end **228b** or may be positioned in casing **206** of secondary wellbore **12a** or within PBR **215** (FIG. **5**) if present.

(159) FIG. **49** illustrates the placement of a whipstock **4976** in the frac window system **226**. The whipstock **4976** may be similar in many respects to the whipstock **3076** of FIG. **30A** above, but for the exclusion of the high-expansion member **3080** (e.g., the high-expansion members are not necessary because of the spacer window sleeve **4010**). The whipstock **4976**, in the illustrated embodiment, is run-in-hole through the low-pressure frac string **4251**.

(160) Turning to FIG. **50**, a straddle stimulation tool **5085** is illustrated extending from the frac window system **226** into the upper secondary wellbore **12b**. In the illustrated embodiment, the straddle stimulation tool **5085** is run-in-hole through the low-pressure frac string **4251**. Straddle stimulation tool **5085** is similar in many respects to the straddle stimulation tool **3285**, but for the exclusion of the high-expansion seals.

(161) FIG. **51** illustrates the removal of the plug **4310**. Those skilled in the art understand the processes that might be used to remove the plug **4310**.

(162) As shown in FIG. **52**, the secondary wellbore **12b** has at least partially been re-stimulated. For example, the secondary wellbore **12b** has been re-stimulated from toe to heel, for example using one or more frac plugs **5210** to isolate the different stimulated regions. In at least one embodiment, as shown, the deployment of the frac plugs **5210** and the re-stimulation process are conducted through the low-pressure frac string **4251**.

(163) FIG. **53** illustrates production from the upper secondary wellbore **12b** or flowback of fluids **5303**, such as hydraulic fracturing fluids and/or hydrocarbons, from fractures **5305** resulting from such an operation, where flowback **5303** from secondary wellbore **12b** is illustrated while secondary wellbore **12a** remains isolated.

(164) FIG. **54** illustrates the removal of the straddle stimulation tool **5085**. In the illustrated embodiment, the straddle stimulation tool **5085** is removed through the low-pressure frac string **4251**.

(165) FIG. **55** illustrates the removal of the deflector **4976**. In the illustrated embodiment, the

deflector **4976** is removed through the low-pressure frac string **4251**.

(166) FIG. **56** illustrates the removal of the plug **4874**. In the illustrated embodiment, the plug **4874** is removed through the low-pressure frac string **4251**. In yet another embodiment, the plug **4874** is drilled out through the low-pressure frac string **4251**. What results is commingled flow from the lower secondary wellbore **12a** and the upper secondary wellbore **12b**. In at least one embodiment, the commingled flow travels to the surface of the wellbore **12** through the frac window system **226** and the low-pressure frac string **4251**. Accordingly, the low-pressure frac string **4251** may ultimately function as production tubing, for at least as long as it remains within the wellbore **12**.

(167) FIG. **57** illustrates the removal of the low-pressure frac string **4251** and the spacer window sleeve **4010**. Those skilled in the art understand the processes that may be used to remove the low-pressure frac string **4251** and the spacer window sleeve **4010**.

(168) Aspects disclosed herein include:

(169) A. A frac window system, the frac window system including: 1) an elongated tubular having a first end and a second end with an opening defined in a wall of the elongated tubular between the first end and the second end, the wall having an inner surface and an outer surface, wherein the opening in the wall is configured to align with a window of a wellbore casing; and 2) a polished bore receptacle coupled to the first end of the elongated tubular, the polished bore receptacle having an inside diameter (ID.sub.1) sufficient to engage with a high-pressure frac string.

(170) B. A well system, the well system including: 1) a first wellbore casing defining an interior annulus and having a window formed there along; 2) a secondary wellbore extending from the window of the first wellbore casing, the first wellbore casing and the secondary wellbore forming a junction; 3) a frac window system disposed within the first wellbore casing at the junction, the frac window system including: a) an elongated tubular having a first end and a second end with an opening defined in a wall of the elongated tubular between the first end and the second end, the wall having an inner surface and an outer surface, wherein the opening in the wall is aligned with the window of the first wellbore casing; and b) a polished bore receptacle coupled to the first end of the elongated tubular, the polished bore receptacle having an inside diameter (ID.sub.1) sufficient to engage with a high-pressure frac string; and 4) a high-pressure frac string coupled to the polished bore receptacle.

(171) C. A wellbore stimulation method, the method including: 1) positioning a frac window system in a first wellbore casing defining an interior annulus and having a window formed there along, the frac window system including: a) an elongated tubular having a first end and a second end with an opening defined in a wall of the elongated tubular between the first end and the second end, the wall having an inner surface and an outer surface; and b) a polished bore receptacle coupled to the first end of the elongated tubular, the polished bore receptacle having an inside diameter (ID.sub.1) sufficient to engage with a high-pressure frac string; and 2) orientating the frac window system so that the opening in the elongated tubular aligns with a junction of a secondary wellbore extending from the cased portion of the first wellbore.

(172) D. A sleeve for use with a frac window system, the sleeve including: 1) a tubular having a first tubular end and a second tubular end; and 2) one or more flow control orifices located in a sidewall of the tubular between the first tubular end and the second tubular end, the tubular configured to be placed within a frac window system at a junction between a first wellbore and a secondary wellbore such that the one or more flow control orifices restrict a flow of wellbore fluid from the secondary wellbore into the tubular.

(173) E. A well system, the well system including: 1) a first wellbore casing defining an interior annulus and having a window formed there along; 2) a secondary wellbore extending from the window of the first wellbore casing, the first wellbore casing and the secondary wellbore forming a junction; 3) a frac window system disposed within the first wellbore casing at the junction, the frac window system including an elongated tubular having a first end and a second end with an opening defined in a wall of the elongated tubular between the first end and the second end, the wall having

an inner surface and an outer surface, wherein the opening in the wall is aligned with the window of the first wellbore casing; and 4) a sleeve positioned within the frac window system, the sleeve including: a) a tubular having a first tubular end and a second tubular end; and b) one or more flow control orifices located in a sidewall of the tubular between the first tubular end and the second tubular end such that the flow control orifices restrict a flow of wellbore fluid from the secondary wellbore into the elongated tubular.

(174) F. A method, the method including: 1) positioning a frac window system in a first wellbore casing defining an interior annulus and having a window formed there along, the frac window system including an elongated tubular having a first end and a second end with an opening defined in a wall of the elongated tubular between the first end and the second end, the wall having an inner surface and an outer surface; 2) orientating the frac window system so that the opening in the elongated tubular aligns with the window and a junction of a secondary wellbore extending from the cased portion of the first wellbore; and 3) positioning a sleeve within the frac window system, the sleeve including: a) a tubular having a first tubular end and a second tubular end; and b) one or more flow control orifices located in a sidewall of the tubular between the first tubular end and the second tubular end such that the flow control orifices restrict a flow of wellbore fluid from the secondary wellbore into the elongated tubular.

(175) G. An isolation sleeve for use with a frac window system, the isolation sleeve including: 1) a tubular having a first tubular end and a second tubular end; and 2) a first high-expansion seal located at least partially along an outer surface of the tubular proximate the first tubular end and a second high-expansion seal located at least partially along the outer surface of the tubular proximate the second tubular end, the first and second high-expansion seals configured to move between a radially retracted state for running the isolation sleeve in hole and a radially expanded state for engaging an inner surface of a frac window system.

(176) H. A well system, the well system including: 1) a first wellbore casing defining an interior annulus and having a window formed there along; 2) a secondary wellbore extending from the window of the first wellbore casing, the first wellbore casing and the secondary wellbore forming a junction; 3) a frac window system disposed within the first wellbore casing at the junction, the frac window system including an elongated tubular having a first end and a second end with an opening defined in a wall of the elongated tubular between the first end and the second end, the wall having an inner surface and an outer surface, wherein the opening in the wall is aligned with the window of the first wellbore casing; and 4) an isolation sleeve positioned within the frac window system, the isolation sleeve including: a) a tubular having a first tubular end and a second tubular end; and b) a first high-expansion seal located at least partially along an outer surface of the tubular proximate the first tubular end and a second high-expansion seal located at least partially along the outer surface of the tubular proximate the second tubular end, the first and second high-expansion seals configured to move between a radially retracted state for running the isolation sleeve in hole and a radially expanded state for engaging the inner surface of the frac window system.

(177) I. A method, the method including: 1) positioning a frac window system in a first wellbore casing defining an interior annulus and having a window formed there along, the frac window system including an elongated tubular having a first end and a second end with an opening defined in a wall of the elongated tubular between the first end and the second end, the wall having an inner surface and an outer surface; 2) orientating the frac window system so that the opening in the elongated tubular aligns with the window and a junction of a secondary wellbore extending from the cased portion of the first wellbore; and 3) positioning an isolation sleeve within the frac window system, the isolation sleeve including: a) a tubular having a first tubular end and a second tubular end; and b) a first high-expansion seal located at least partially along an outer surface of the tubular proximate the first tubular end and a second high-expansion seal located at least partially along the outer surface of the tubular proximate the second tubular end, the first and second high-expansion seals configured to move between a radially retracted state for running the isolation

sleeve in hole and a radially expanded state for engaging the inner surface of the frac window system; and 4) moving the first and second high-expansion seals from the radially retracted state to the radially expanded state to engage the inner surface of the frac window system.

(178) J. A whipstock for use with a frac window system, the whipstock including: 1) a housing having a first whipstock end with a contoured surface and a second whipstock end; and 2) one or more high-expansion members located at least partially along an outer surface of the housing the one or more high-expansion members configured to move between a radially retracted state for running the whipstock in hole and a radially expanded state for engaging an inner surface of a frac window system.

(179) K. A well system, the well system including: 1) a first wellbore casing defining an interior annulus and having a window formed there along; 2) a secondary wellbore extending from the window of the first wellbore casing, the first wellbore casing and the secondary wellbore forming a junction; 3) a frac window system disposed within the first wellbore casing at the junction, the frac window system including an elongated tubular having a first end and a second end with an opening defined in a wall of the elongated tubular between the first end and the second end, the wall having an inner surface and an outer surface, wherein the opening in the wall is aligned with the window of the first wellbore casing; and 4) a whipstock positioned within the frac window system, the whipstock including: a) a housing having a first whipstock end with a contoured surface and a second whipstock end; and b) one or more high-expansion members located at least partially along an outer surface of the housing the one or more high-expansion members configured to move between a radially retracted state for running the whipstock in hole and a radially expanded state for engaging an inner surface of a frac window system.

(180) L. A method, the method including: 1) positioning a frac window system in a first wellbore casing defining an interior annulus and having a window formed there along, the frac window system including an elongated tubular having a first end and a second end with an opening defined in a wall of the elongated tubular between the first end and the second end, the wall having an inner surface and an outer surface; 2) orientating the frac window system so that the opening in the elongated tubular aligns with the window and a junction of a secondary wellbore extending from the cased portion of the first wellbore; and 3) positioning a whipstock within the frac window system, the whipstock including: a) a housing having a first whipstock end with a contoured surface and a second whipstock end; and b) one or more high-expansion members located at least partially along an outer surface of the housing the one or more high-expansion members configured to move between a radially retracted state for running the whipstock in hole and a radially expanded state for engaging an inner surface of a frac window system; and 4) moving the one or more high-expansion members from the radially retracted state to the radially expanded state to engage the inner surface of the frac window system.

(181) M. A frac window system, the frac window system including: 1) an elongated tubular having a first end and a second end with an opening defined in a wall of the elongated tubular between the first end and the second end, the wall having an inner surface and an outer surface, wherein the opening in the wall is configured to align with a window of a first wellbore casing; and 2) a spacer window sleeve positioned within the elongated tubular, the spacer window sleeve including a tubular having a first tubular end and a second tubular end with a second opening defined in a second wall of the tubular between the first tubular end and the second tubular end, the second wall having a second inner surface and a second outer surface, wherein the second opening in the second wall is configured to at least partially align with the opening in the wall of the elongated tubular.

(182) N. A well system, the well system including: 1) a first wellbore casing defining an interior annulus and having a window formed there along; 2) a secondary wellbore extending from the window of the first wellbore casing, the first wellbore casing and the secondary wellbore forming a junction; 3) a frac window system disposed within the first wellbore casing at the junction, the frac

window system including: a) an elongated tubular having a first end and a second end with an opening defined in a wall of the elongated tubular between the first end and the second end, the wall having an inner surface and an outer surface, wherein the opening in the wall is configured to align with a window of a first wellbore casing; and b) a spacer window sleeve positioned within the elongated tubular, the spacer window sleeve including a tubular having a first tubular end and a second tubular end with a second opening defined in a second wall of the tubular between the first tubular end and the second tubular end, the second wall having a second inner surface and a second outer surface, wherein the second opening in the second wall is configured to at least partially align with the opening in the wall of the elongated tubular.

(183) O. A wellbore stimulation method, the method including: positioning a frac window system in a first wellbore casing defining an interior annulus and having a window formed there along, the frac window system including: 1) an elongated tubular having a first end and a second end with an opening defined in a wall of the elongated tubular between the first end and the second end, the wall having an inner surface and an outer surface, wherein the opening in the wall is configured to align with a window of a first wellbore casing; and 2) a spacer window sleeve positioned within the elongated tubular, the spacer window sleeve including a tubular having a first tubular end and a second tubular end with a second opening defined in a second wall of the tubular between the first tubular end and the second tubular end, the second wall having a second inner surface and a second outer surface, wherein the second opening in the second wall is configured to at least partially align with the opening in the wall of the elongated tubular.

(184) Aspects A, B, C, D, E, F, G, H, I, J, K, L, M, N, and O may have one or more of the following additional elements in combination: Element 1: wherein the elongated tubular has an inside diameter (ID.sub.3), and further wherein the inside diameter (ID.sub.1) of the polished bore receptacle is greater than the inside diameter (ID.sub.3) of the elongated tubular. Element 2: wherein the inside diameter (ID.sub.1) of the polished bore receptacle is at least 5½". Element 3: wherein the inside diameter (ID.sub.1) of the polished bore receptacle is at least 7". Element 4: wherein the polished bore receptacle is capable of withstanding the stresses of stimulating at a pressure of above 5,000-psi. Element 5: wherein the polished bore receptacle is capable of withstanding the stresses of stimulating at a pressure of at least 10,000-psi. Element 6: wherein the polished bore receptacle is capable of withstanding the stresses of stimulating at a pressure of over 12,500-psi. Element 7: wherein the polished bore receptacle is capable of withstanding the stresses of stimulating at a pressure of 15,000-psi. Element 8: wherein a length (L.sub.1) of the polished bore receptacle is at least two times the inside diameter (ID.sub.1) of the polished bore receptacle. Element 9: wherein the polished bore receptacle comprises at least a 125-ksi grade material. Element 10: wherein the high-pressure frac string is an extremely high-pressure frac string. Element 11: wherein the high-pressure frac string is a super high-pressure frac string. Element 12: wherein the high-pressure frac string has an inside diameter (ID.sub.5) of at least 5½". Element 13: further including a whipstock positioned within the frac window system proximate the junction. Element 14: further including a straddle stimulation tool extending through the window into the secondary wellbore. Element 15: further including one or more debris barrier layers located proximate an uphole surface of the straddle stimulation tool. Element 16: wherein the straddle stimulation tool has one or more sealing elements along its outer surface, and further including one or more protective sheaths at least partially covering the one or more sealing elements. Element 17: further including coupling a high-pressure frac string with the polished bore receptacle. Element 18: further including stimulating the secondary wellbore through the high-pressure frac string. Element 19: further including stimulating the first wellbore through the high-pressure frac string. Element 20: further including passing a wellbore device having an outside diameter (OD) of at least 5½" through the high-pressure frac string and into at least one of the first wellbore or the secondary wellbore. Element 21: wherein the wellbore device is a wellbore plug. Element 22: wherein the wellbore device is an isolation sleeve. Element 23: wherein the wellbore device is a

Element 24: wherein the wellbore device is a straddle stimulation tool. Element 25: wherein the wellbore device is a frac plug. Element 26: further including two or more flow control orifices located in the sidewall of the tubular between the first tubular end and the second tubular end. Element 27: further including an uphole seal located at least partially along an outer surface of the tubular proximate the first tubular end and a downhole seal located at least partially along the outer surface of the tubular proximate the second tubular end. Element 28: wherein the uphole seal is a first uphole seal and the downhole seal is a first downhole seal, and further including a second uphole seal located at least partially along the outer surface of the tubular proximate the first tubular end and a second downhole seal located at least partially along the outer surface of the tubular proximate the second tubular end. Element 29: further including a depth mechanism disposed at least partially along the outer surface of the tubular, the depth mechanism configured to engage a related depth mechanism disposed along an inner surface of a frac window system. Element 30: wherein the depth mechanism is positioned between the uphole seal and the first tubular end. Element 31: wherein an outside diameter (OD.sub.s) of the tubular is at least 5.5". Element 32: wherein the one or more flow control orifices are located proximate the junction. Element 33: wherein the one or more flow control orifices are located between opposing edges of the window. Element 34: wherein positioning the sleeve includes positioning the sleeve such that the one or more flow control orifices are located proximate the junction. Element 35: wherein positioning the sleeve includes positioning the sleeve such that the one or more flow control orifices are located between opposing edges of the window. Element 36: further including removing the sleeve and positioning a second sleeve within the frac window system, the second sleeve having a different number or size of one or more second flow control orifices than the sleeve. Element 37: further including a depth mechanism disposed at least partially along the outer surface of the tubular, the depth mechanism configured to engage a related depth mechanism disposed along an inner surface of the frac window system. Element 38: wherein the depth mechanism is positioned between the first high-expansion seal and the first tubular end. Element 39: wherein the first high-expansion seal and the second high-expansion seal are spaced such that they span a junction between a first wellbore and a secondary wellbore that the frac window system is located. Element 40: wherein the first and second seals are located in the radially retracted state such that they do not engage the inner surface of the frac window system. Element 41: wherein the first and second seals are located in the radially expanded state such that they engage the inner surface of the frac window system. Element 42: wherein the first and second high-expansion seals can seal a frac pressure of at least 5,000-psi. Element 43: wherein the first and second high-expansion seals can seal a frac pressure of at least 10,000-psi. Element 44: wherein the first and second high-expansion seals can seal a frac pressure of at least 12,500-psi. Element 45: further including stimulating a first wellbore associated with the first wellbore casing through the isolation sleeve while the isolation sleeve isolates the secondary wellbore. Element 46: wherein stimulating includes stimulating with a frac pressure of at least 5,000-psi. Element 47: wherein stimulating includes stimulating with a frac pressure of at least 10,000-psi. Element 48: wherein stimulating includes stimulating with a frac pressure of at least 12,500-psi. Element 49: wherein the one or more high-expansion members are one or more scissor type high-expansion members. Element 50: wherein the one or more high-expansion members each have a plurality of teeth for engaging the inner surface of the frac window system. Element 51: wherein the whipstock is a neckless whipstock. Element 52: wherein the whipstock is necked or extended necked whipstock. Element 53: further including a depth mechanism disposed at least partially along the outer surface of the housing, the depth mechanism configured to engage a related depth mechanism disposed along an inner surface of the frac window system. Element 54: wherein the depth mechanism is positioned between the one or more high-expansion members and the second whipstock end. Element 55: wherein the one or more high-expansion members are located in the radially retracted state such that they do not engage the inner surface of the frac window system. Element 56: wherein the one

or more high-expansion members are located in the radially expanded state such that they engage the inner surface of the frac window system. Element 57: wherein the spacer window sleeve further includes an orientation device. Element 58: wherein the orientation device is a muleshoe. Element 59: wherein the muleshoe is located proximate the first tubular end. Element 60: wherein the muleshoe is located between the second opening and the second tubular end. Element 61: wherein the spacer window sleeve further includes an uphole seal located at least partially along an outer surface of the tubular proximate the first tubular end and a downhole seal located at least partially along the outer surface of the tubular proximate the second tubular end. Element 62: wherein the uphole seal is a first uphole seal and the downhole seal is a first downhole seal, and further including a second uphole seal located at least partially along the outer surface of the tubular proximate the first tubular end and a second downhole seal located at least partially along the outer surface of the tubular proximate the second tubular end. Element 63: further including an isolation sleeve positioned within the spacer window sleeve. Element 64: further including orientating the frac window system so that the opening in the elongated tubular aligns with a junction of a secondary wellbore extending from the cased portion of the first wellbore.

(185) Those skilled in the art to which this application relates will appreciate that other and further additions, deletions, substitutions, and modifications may be made to the described embodiments.

Claims

1. A well system, comprising: a first wellbore casing defining an interior annulus and having a window formed there along; a secondary wellbore extending from the window of the first wellbore casing, the first wellbore casing and the secondary wellbore forming a junction; a frac window system disposed within the first wellbore casing at the junction, the frac window system including: an elongated tubular having a first end and a second end with an opening defined in a wall of the elongated tubular between the first end and the second end, the wall having an inner surface and an outer surface, wherein the opening in the wall is aligned with the window of the first wellbore casing; and a polished bore receptacle coupled to the first end of the elongated tubular, the polished bore receptacle having a length (L1) and an inside diameter (ID.sub.1) sufficient to engage with one or more seals of a high-pressure frac string, such that the elongated tubular may be subjected to fluid pressures employed from the high-pressure frac string to frac the secondary wellbore exposed by the window of the wellbore casing; and a high-pressure frac string coupled to the polished bore receptacle with the one or more seals.
2. The well system as recited in claim 1, wherein the elongated tubular has an inside diameter (ID 3), and further wherein the inside diameter (ID.sub.1) of the polished bore receptacle is greater than the inside diameter (ID.sub.3) of the elongated tubular.
3. The well system as recited in claim 1, wherein the inside diameter (ID 1) of the polished bore receptacle is at least 5½".
4. The well system as recited in claim 1, wherein the inside diameter (ID 1) of the polished bore receptacle is at least 7".
5. The well system as recited in claim 1, wherein the polished bore receptacle is capable of withstanding the stresses of stimulating at a pressure of above 5,000-psi.
6. The well system as recited in claim 1, wherein the polished bore receptacle is capable of withstanding the stresses of stimulating at a pressure of at least 10,000-psi.
7. The well system as recited in claim 1, wherein the polished bore receptacle is capable of withstanding the stresses of stimulating at a pressure of over 12,500-psi.
8. The well system as recited in claim 1, wherein the polished bore receptacle is capable of withstanding the stresses of stimulating at a pressure of 15,000-psi.
9. The well system as recited in claim 1, wherein a length (L.sub.1) of the polished bore receptacle is at least two times the inside diameter (ID.sub.1) of the polished bore receptacle.

10. The well system as recited in claim 1, wherein the polished bore receptacle comprises at least a 125-ksi grade material.
 11. The well system as recited in claim 1, wherein the high-pressure frac string is an extremely high-pressure frac string.
 12. The well system as recited in claim 1, wherein the high-pressure frac string is a super high-pressure frac string.
 13. The well system as recited in claim 1, wherein the high-pressure frac string has an inside diameter (ID 5) of at least 5½".
 14. The well system as recited in claim 1, further including a whipstock positioned within the frac window system proximate the junction.
 15. The well system as recited in claim 14, further including a straddle stimulation tool extending through the window into the secondary wellbore.
 16. The well system as recited in claim 15, further including one or more debris barrier layers located proximate an uphole surface of the straddle stimulation tool.
 17. The well system as recited in claim 15, wherein the straddle stimulation tool has one or more sealing elements along its outer surface, and further including one or more protective sheaths at least partially covering the one or more sealing elements.
 18. A wellbore stimulation method, comprising: positioning a frac window system in a first wellbore casing defining an interior annulus and having a window formed there along, the frac window system including: an elongated tubular having a first end and a second end with an opening defined in a wall of the elongated tubular between the first end and the second end, the wall having an inner surface and an outer surface; and a polished bore receptacle coupled to the first end of the elongated tubular, the polished bore receptacle having a length (L1) and an inside diameter (ID.sub.1) sufficient to engage with one or more seals of a high-pressure frac string, such that the elongated tubular may be subjected to fluid pressures employed from the high-pressure frac string to frac a secondary wellbore exposed by the window of the wellbore casing; and orientating the frac window system so that the opening in the elongated tubular aligns with a junction of the secondary wellbore extending from the cased portion of the first wellbore, wherein a high-pressure frac string is coupled with the polished bore receptacle.
 19. The wellbore stimulation method as recited in claim claim 18, further including stimulating the secondary wellbore through the high-pressure frac string.
 20. The wellbore stimulation method as recited in claim claim 18, further including stimulating the first wellbore through the high-pressure frac string.
 21. The wellbore stimulation method as recited in claim claim 18, further including passing a wellbore device having an outside diameter (OD) of at least 5½" through the high-pressure frac string and into at least one of the first wellbore or the secondary wellbore.
 22. The wellbore stimulation method as recited in claim 21, wherein the wellbore device is a wellbore plug.
 23. The wellbore stimulation method as recited in claim 21, wherein the wellbore device is an isolation sleeve.
 24. The wellbore stimulation method as recited in claim 21, wherein the wellbore device is a whipstock.
 25. The wellbore stimulation method as recited in claim 21, wherein the wellbore device is a straddle stimulation tool.
 26. The wellbore stimulation method as recited in claim 21, wherein the wellbore device is a frac plug.
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